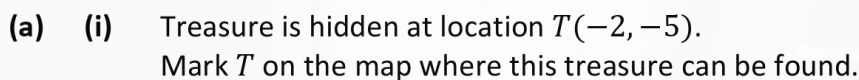
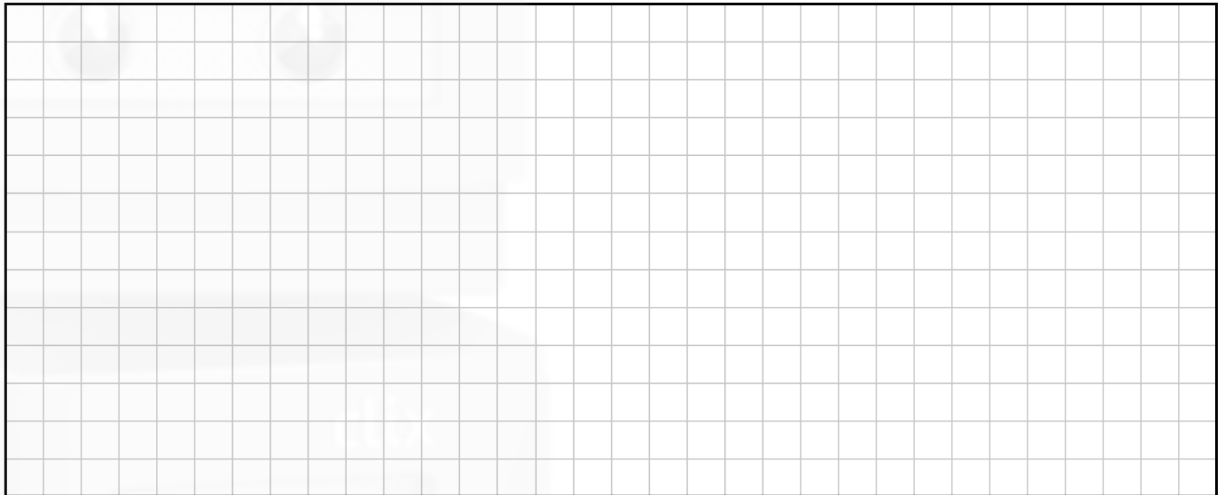


A TY maths class has created a game involving a co-ordinate treasure map, as shown below. The game consists of tasks that involve directions, distances, and locations. The tasks in the game are based on the map.

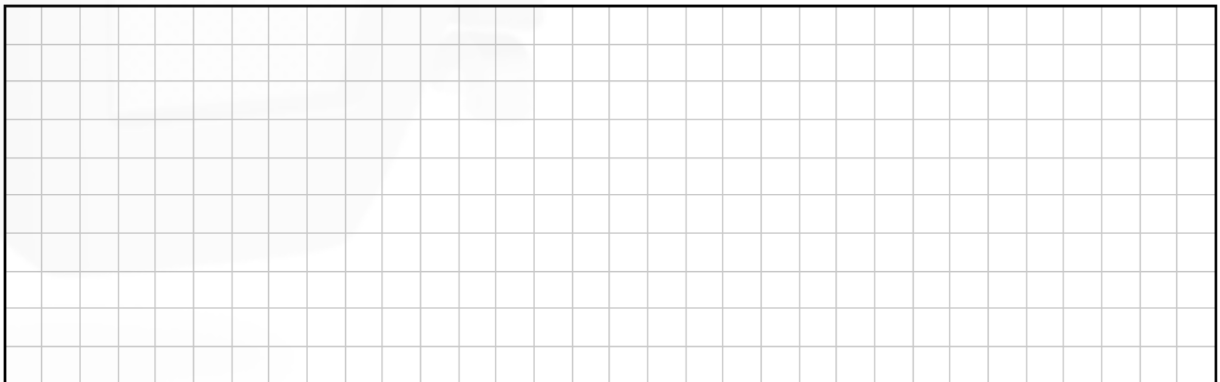
[illegible]

- [illegible]

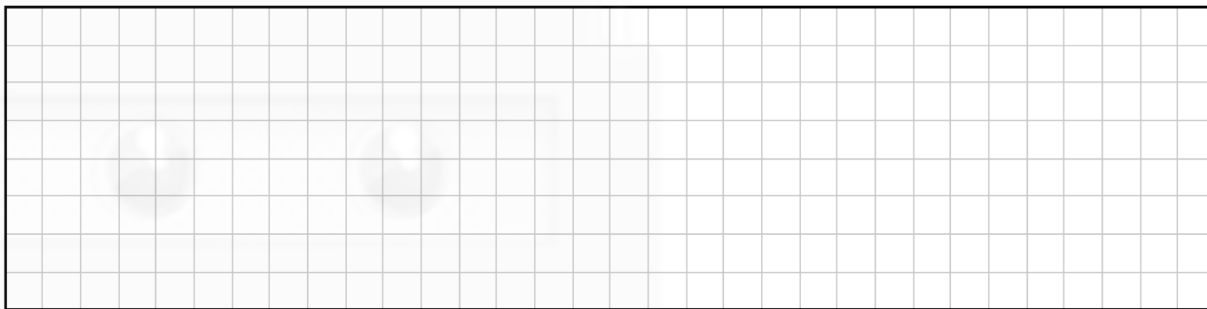
- (b) (i) A clue to another treasure is hidden in a locked box at point $B(6, 4)$.
The 4-digit code to open the box is d^4 , where d is the distance from B to $C(-3, 2)$,
and $d \in \mathbb{N}$.
Find the 4-digit code (d^4).



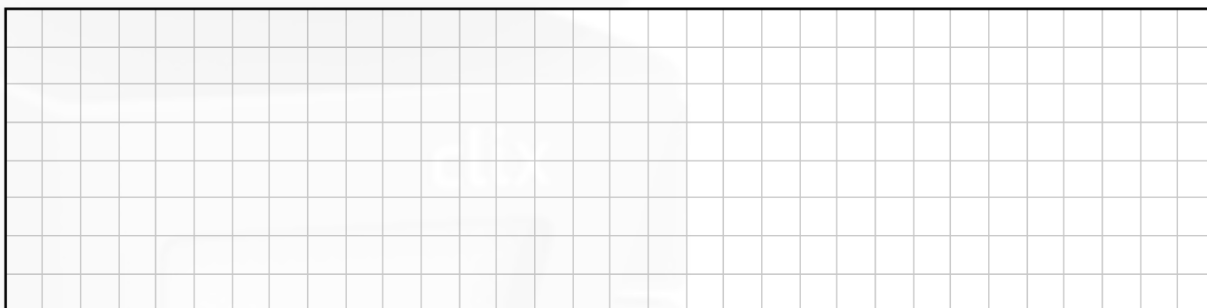
- (ii) A meeting point $P(2, m)$ is below the x -axis.
 P is a distance of $\sqrt{41}$ from point $B(6, 4)$.
Find the value of m **and** plot P on the map above.



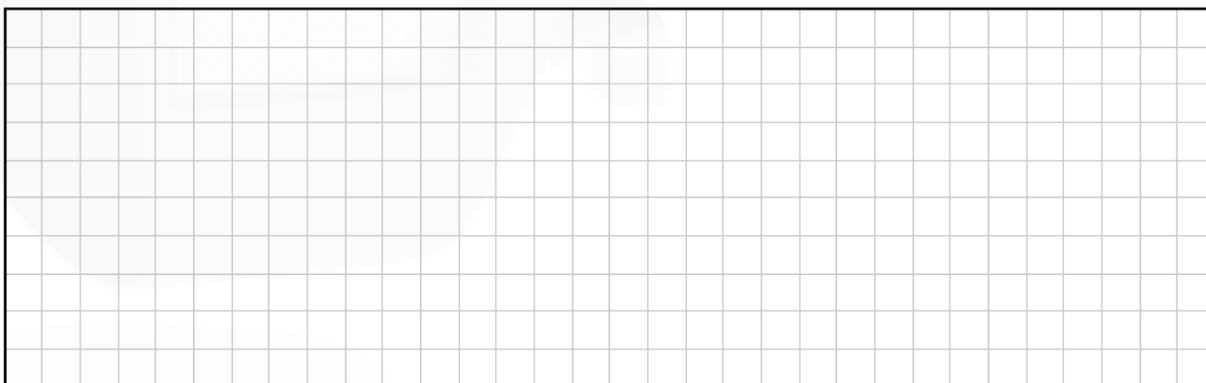
- (c) (i) The line k has equation $x - y - 3 = 0$.
Verify, using substitution, that the point $T(-2, -5)$ is on k .



- (ii) Another treasure also needs be somewhere on the line k .
You must pick a spot along k to contain this treasure.
Use algebra to find another point on k , other than T .



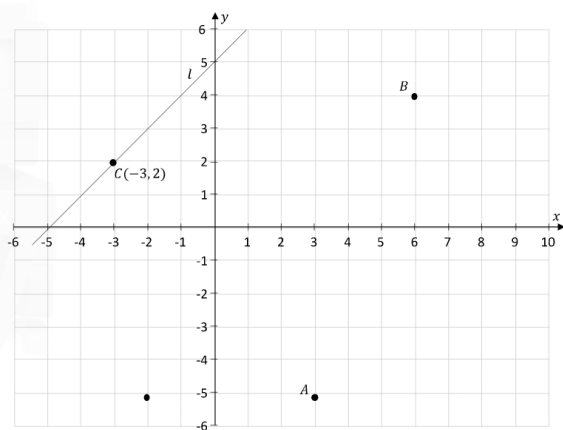
- (iii) A spade for digging is hidden on line l which is parallel to k .
The line l contains the point $C(-3, 2)$.
Find the equation of line l .



Marking Scheme

(a)

(i)



Scale 10B (0, 5, 10)

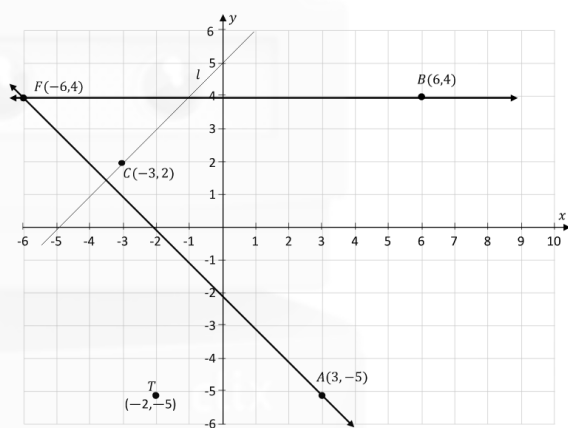
Partial Credit:

- Point plotted as $(-5, -2)$
- One or both coordinates plotted as a + value

(a)

(ii)

$$F(-6, 4)$$



or

$$y - (-5) = -1(x - 3)$$

$$x + y = -2$$

$$y - 4 = 0(x - 6)$$

$$y = 4$$

$$\Rightarrow x + 4 = -2$$

$$x = -6$$

$$F = (-6, 4)$$

(b)

(i)

$$d = \sqrt{81 + 4}$$

$$= \sqrt{85}$$

$$= 9(\in \mathbb{N})$$

$$\Rightarrow d^4 = 6561$$

Note:

Accept $d = \sqrt{85}$, $d^4 = 7225$
for Full Credit

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Slope = $\frac{\text{rise}}{\text{run}}$
- A or B labelled correctly
- Equation of line with some correct substitution
- One line correctly drawn through A or B

High Partial Credit:

Significant Work

- Both lines correctly drawn, F not identified
- Both equations found but not solved

Scale 10C (0, 3, 7, 10)

Low Partial Credit:

Work of merit

- Some correct substitution into formula to find d

High Partial Credit:

Significant Work

- Formula fully substituted
- Correct answer without work

(b)

(ii)

$$|PB| = \sqrt{16 + (m - 4)^2} = \sqrt{41}$$

$$16 + m^2 - 8m + 16 = 41$$

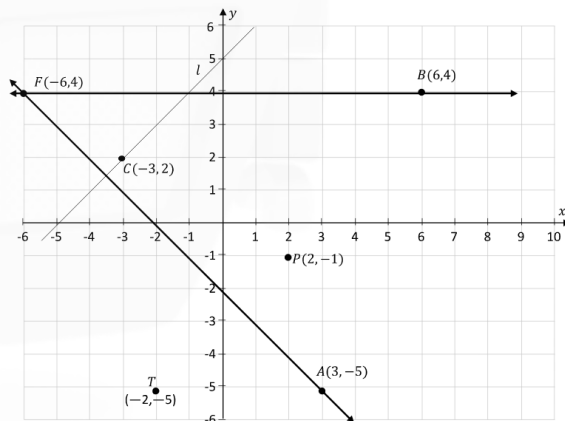
$$m^2 - 8m - 9 = 0$$

$$(m + 1)(m - 9) = 0$$

$$\Rightarrow m = -1$$

($m = 9$ not valid)

$P(2, -1)$ (plot on map)



(c)

(i)

$$k: x - y - 3 = 0$$

$$T \in k \Rightarrow (-2) - (-5) - 3 = 0$$

$$-2 + 5 - 3 = 0$$

$$[5 - 5 = 0]$$

$$[0 = 0 \text{ True}]$$

Scale 5D (0, 2, 3, 4, 5)

Low Partial Credit:

Work of merit

- Some substitution into Pythagoras or distance formula to find $|PB|$

Mid Partial Credit:

- Formula fully substituted
- Incorrect substitution into formula followed by consistent value of m

High Partial Credit:

Significant Work

- Works to the quadratic equation
- One relevant value of m identified but P not plotted

Note:

If 2 values of m given but only 1 point plotted award Full Credit

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Some correct substitution into equation of line k

High Partial Credit:

Significant Work

- Full correct substitution into equation of line k

(c)
(ii)

Example:

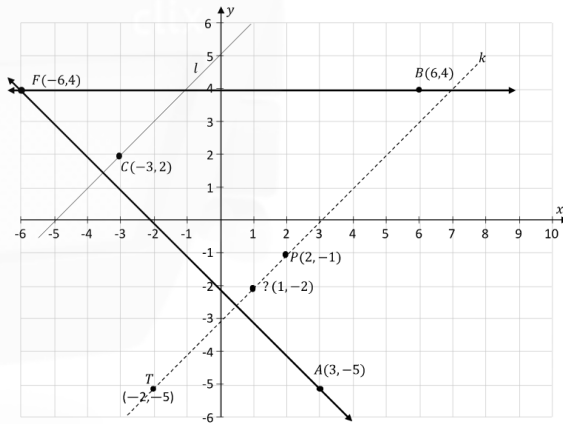
$$k: x - y - 3 = 0$$

$$\text{Let } x = 1$$

$$\Rightarrow 1 - y - 3 = 0$$

$$y = -2$$

$$(1, -2) \in k$$



Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Random selection of a domain value to find a point $(x, y) \in k$
- Graphical solution

High Partial Credit:

Significant Work

- Correct calculation of the corresponding range value

Note: Correct substitution of any point that is shown to satisfy the equation of line k merits Full Credit

(c)
(iii)

$$k: x - y - 3 = 0$$

$$y = x - 3$$

$$\Rightarrow m_l = 1$$

$$\Rightarrow m_k = 1$$

Eqn of l :

$$y - 2 = 1[x - (-3)]$$

$$y - 2 = x + 3$$

or

$$y = x + 5$$

or

$$l: x - y + 5 = 0$$

Scale10C (0, 3, 7, 10)

Low Partial Credit:

Work of merit

- Effort at finding slope of k or l
- Relevant work on diagram
- Slope = $\frac{\text{rise}}{\text{run}}$

High Partial Credit:

Significant Work

- Equation of line formula with full substitution

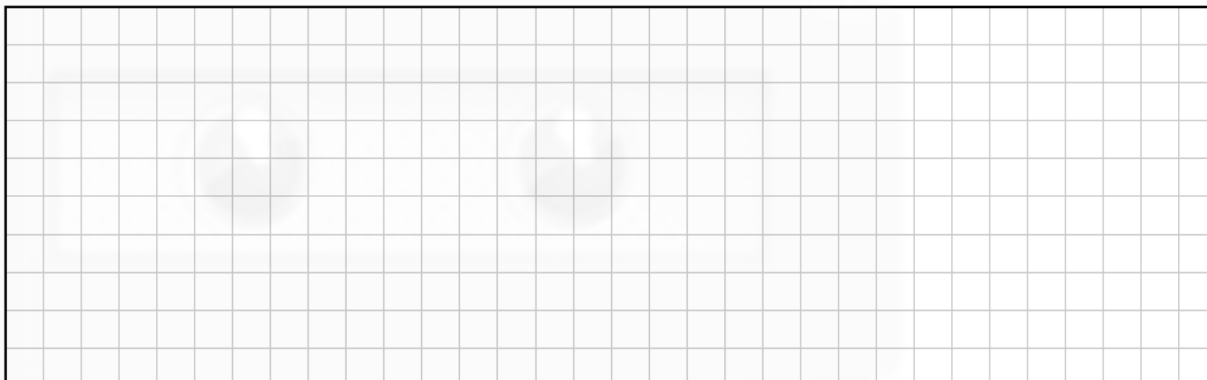
Note: Accept

$$y - 2 = 1(x + 3) \text{ for Full Credit}$$

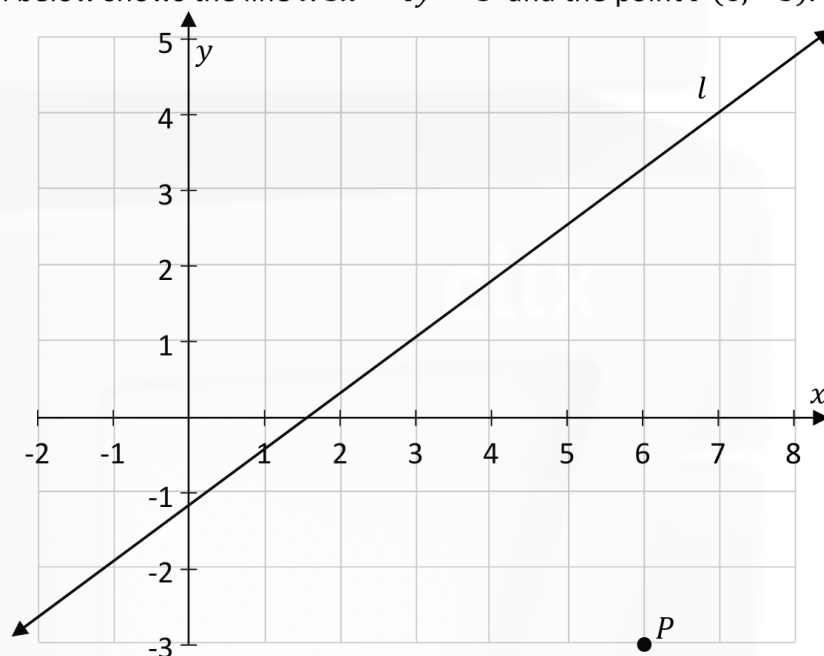
Question 3

(30 marks)

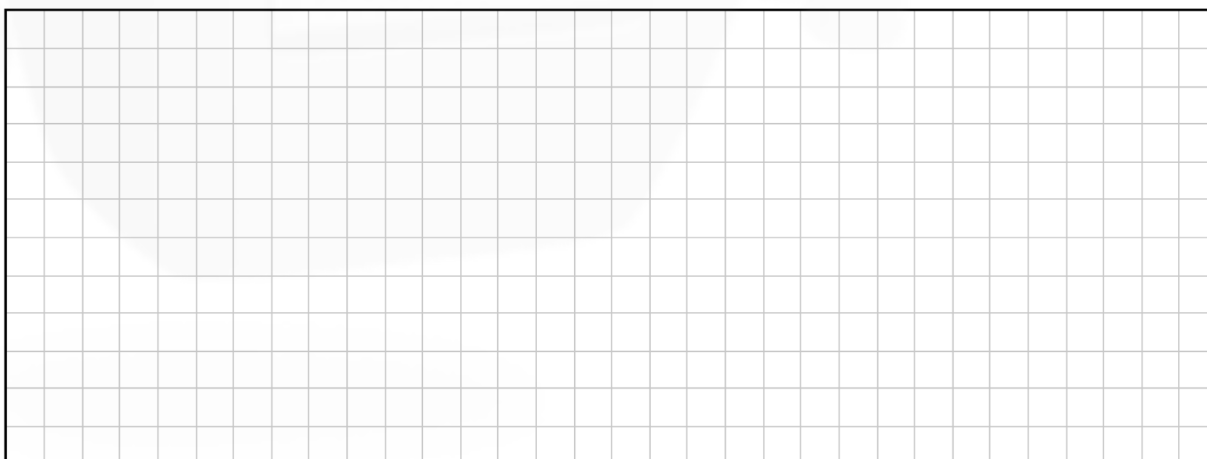
- (a) A line n passes through the points $A(-1, 2)$ and $B(0, -2)$.
Write the equation of n in the form $y = mx + c$, where $m, c \in \mathbb{Z}$.



- (b) The diagram below shows the line $l: 3x - 4y = 5$ and the point $P(6, -3)$.



- (i) Find the equation of the line k through the point P that is perpendicular to the line l .
Write your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.



(ii) Find the point of intersection of the lines $l: 3x - 4y = 5$ and $h: 2x - y = 10$.



(a)

$$\text{slope} = \frac{-2 - 2}{0 - (-1)} = -4$$

$$y - 2 = -4[x - (-1)]$$

$$y = -4x - 4 + 2$$

$$y = -4x - 2$$

or

$$A: 2 = -m + c$$

$$B: -2 = m(0) + c$$

$$\Rightarrow c = -2$$

$$\text{From A: } 2 = -m - 2$$

$$\Rightarrow m = -4$$

$$\rightarrow y = -4x - 2$$

(b)

(i)

$$\text{slope of } l = \frac{3}{4}$$

$$\text{perpendicular slope} = \frac{-4}{3}$$

$$y - (-3) = \frac{-4}{3}(x - 6)$$

$$y + 3 = \frac{-4}{3}(x - 6)$$

$$k: 4x + 3y - 15 = 0$$

Scale 10D (0, 3, 5, 8, 10)

Low Partial Credit:

Work of merit

- Slope formula with some substitution

Mid Partial Credit:

- m found

High Partial Credit:

Significant Work

- Eq. of line formula fully substituted
- One error and finishes correctly
- Answer not in required format

or

Scale 10D (0, 3, 5, 8, 10)

Low Partial Credit:

- Work of merit
- Some correct substitution into equation $y = mx + c$

Mid Partial Credit:

- Full substitution into both A & B
- m or c found

High Partial Credit:

Significant Work

- m and c found
- One error and finishes correctly
- Answer not in required format

Scale 10D (0, 3, 5, 8, 10)

Low Partial Credit:

Work of merit

- slope of $l = \frac{3}{4}$
- Eq. of line formula with some substitution

Mid Partial credit:

$$\text{slope of } k = \frac{-4}{3}$$

High Partial credit:

Significant Work

- Eq. of line formula fully substituted
- One error and finishes correctly
- Answer not in required format

(b)

(ii)

$$\begin{array}{rcl} l: & 3x - 4y = 5 \\ h: & -8x + 4y = -40 \\ \hline & -5x = -35 \\ & x = 7 \end{array}$$

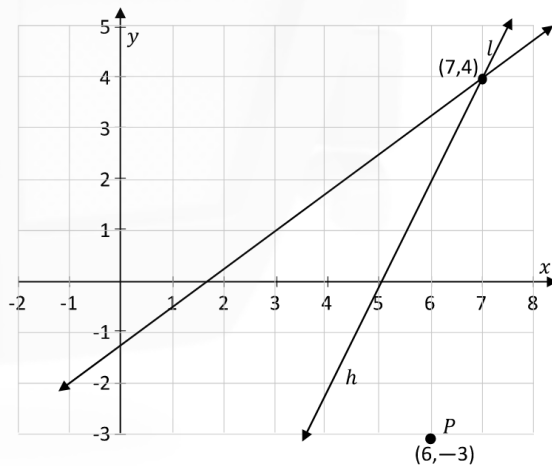
$$\begin{aligned} \Rightarrow -8(7) + 4y &= -40 \\ 4y &= 56 - 40 \\ 4y &= 16 \\ y &= 4 \end{aligned}$$

$$l \cap k = (7, 4)$$

or

Graphical Method

Line h : (5, 0), (6, 2) or (3, -4)



Scale 10C (0, 3, 7, 10)

Low Partial Credit:

Work of merit

- Effort to equate coefficients of x or y
- Effort at isolating one variable
- Effort at drawing line h
- Correct answer without work

High Partial Credit:

Significant Work

- One variable found
- One error and finishes correctly

or

Scale 10C (0, 3, 7, 10)

Low Partial Credit:

Work of merit

- Finds one point on h

High Partial Credit:

Significant Work

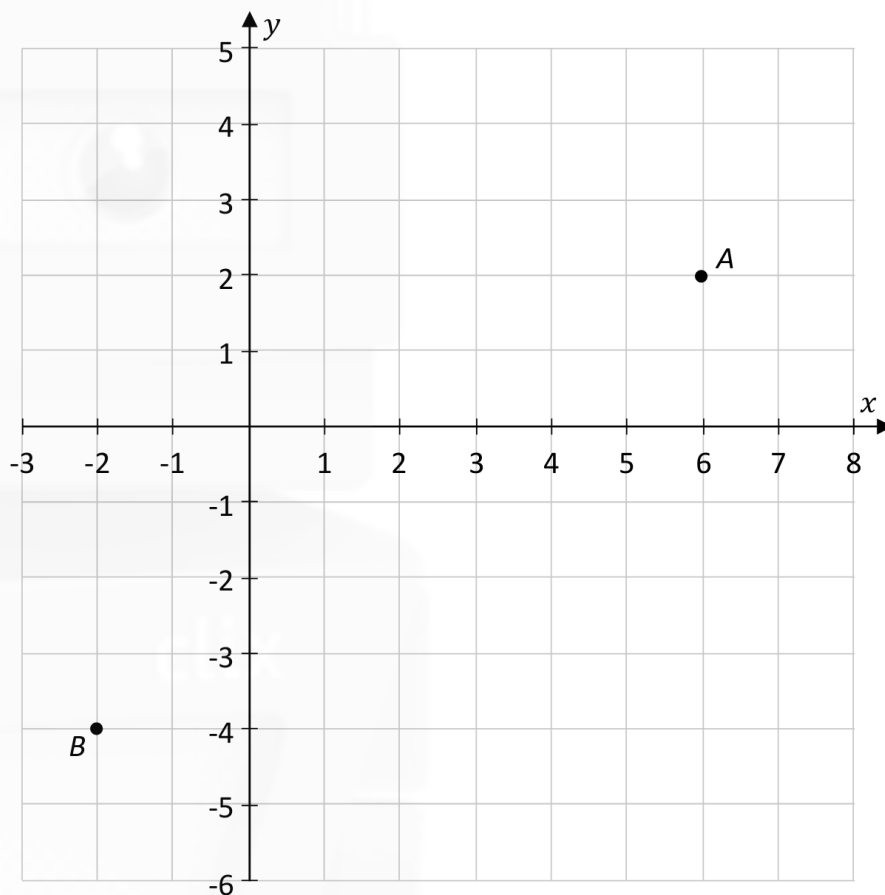
- Draws h

Full Credit:

- $l \cap k = (7, 4)$

(30 marks)

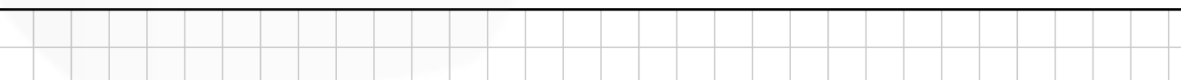
The co-ordinate diagram below shows two points A and B .



- (a) (i)** Write down the co-ordinates of A and of B .

$$A = \left(\begin{array}{cc} & \\ & \end{array} \right) \qquad B = \left(\begin{array}{cc} & \\ & \end{array} \right)$$

- (ii) Find the co-ordinates of the midpoint of $[AB]$.



(a)

(i)

$A(6, 2)$ and $B(-2, -4)$

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- One correct co-ordinate
- Ordinates reversed

High Partial Credit:

Significant Work

- One point correct

Full Credit:

- Correct answer

(a)

(ii)

$$\left[\frac{6 + (-2)}{2}, \frac{2 + (-4)}{2} \right]$$

$(2, -1)$

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Some correct substitution into midpoint formula

High Partial Credit:

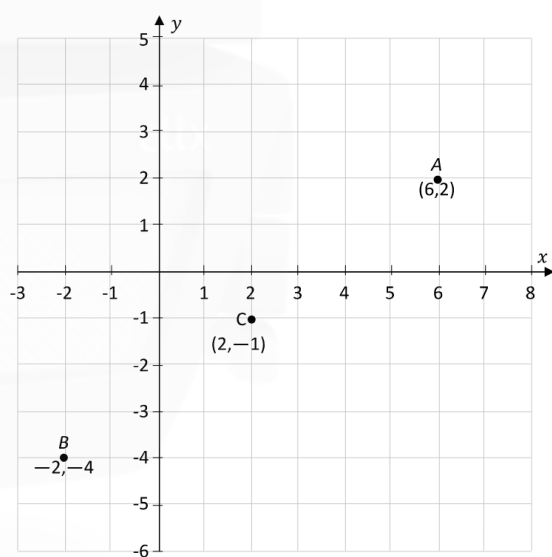
Significant Work

- Full substitution into midpoint formula
- One error and finishes correctly

Full Credit:

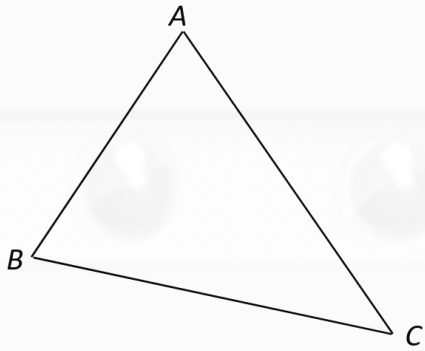
- Correct answer without supporting work

Graphical Solution:

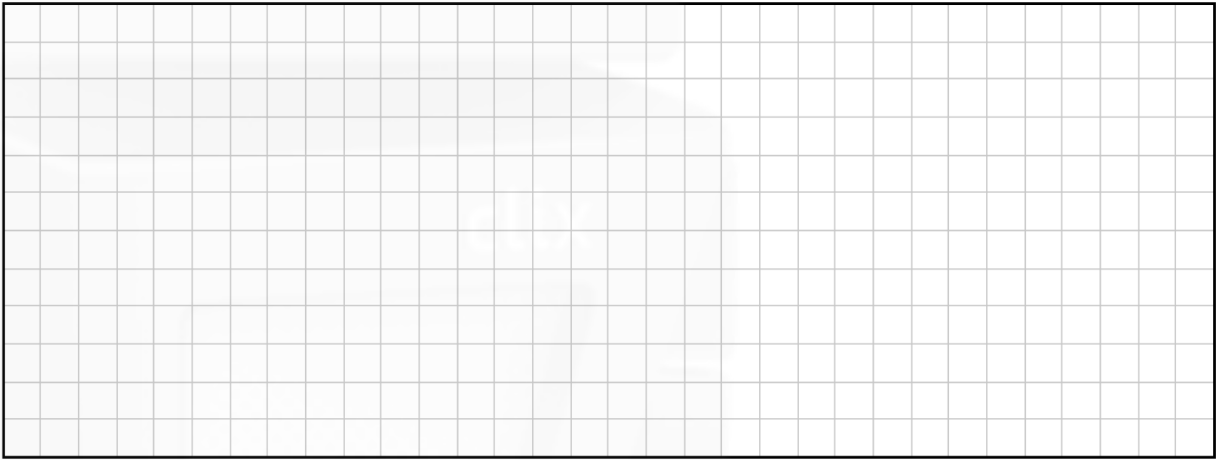


Question 2**(25 marks)**

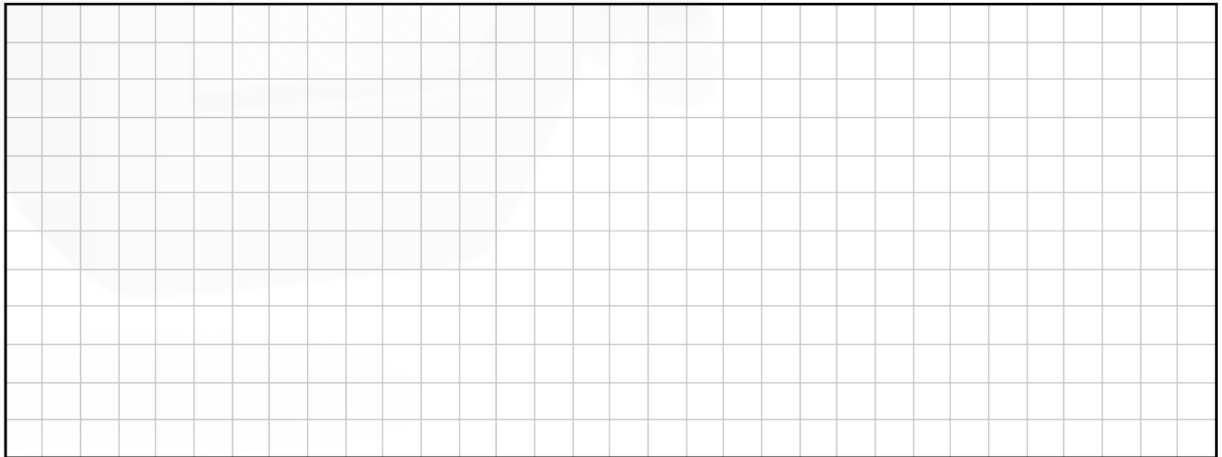
The points $A(4, 6)$, $B(-2, 2)$ and $C(10, 0)$ are the vertices of the triangle ABC shown below.



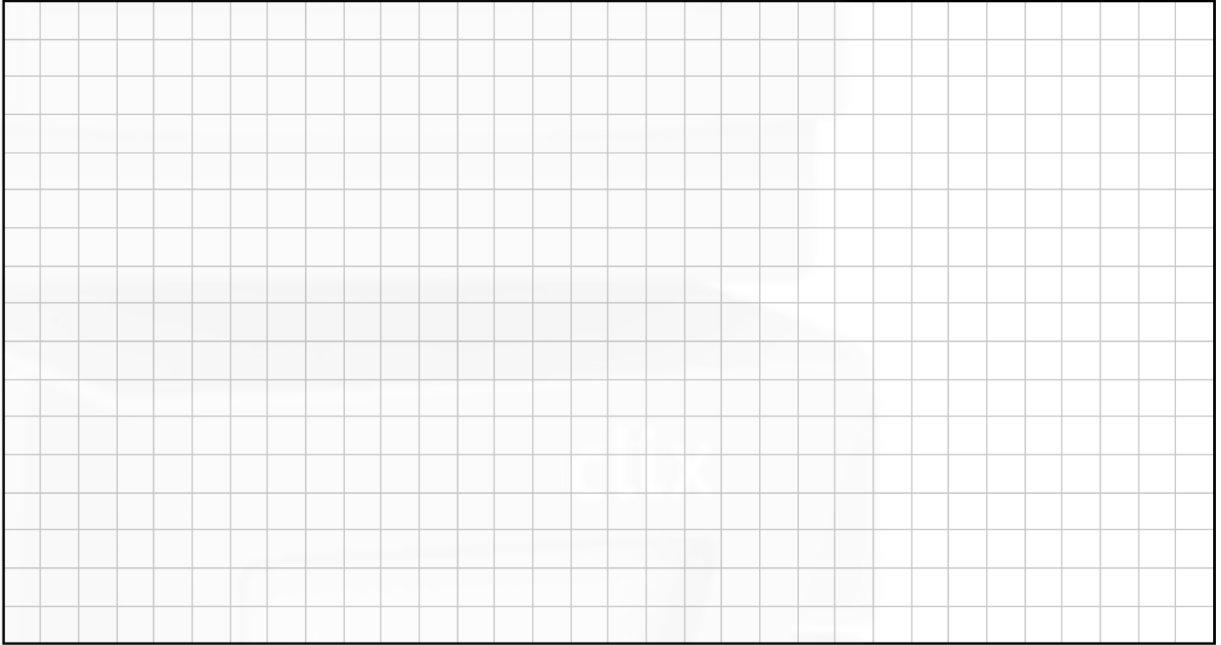
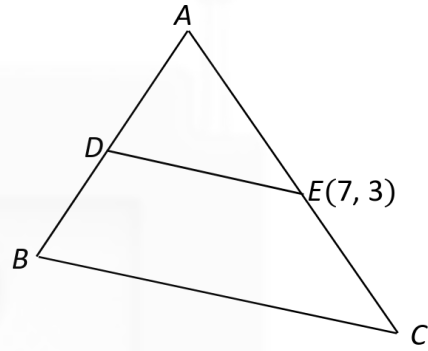
- (a) Find $|AB|$, the length of $[AB]$. Give your answer in the form $a\sqrt{b}$ units, where $a, b \in \mathbb{N}$.



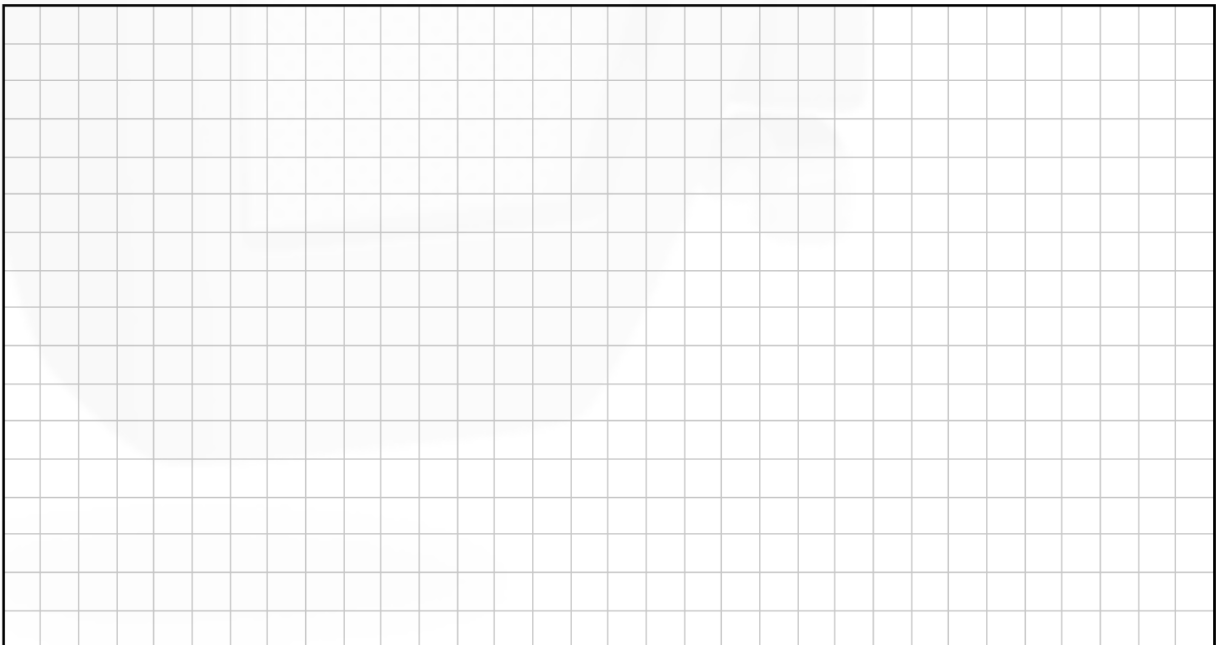
- (b) (i) Find the coordinates of D , the midpoint of $[AB]$.



- (ii) In the triangle ABC , the point $E(7, 3)$ is the midpoint of $[AC]$.
Show that DE is parallel to BC .



- (c) Find the area of the triangle ABC .



Q2	Model Solution – 25 Marks	Marking Notes
(a)	$ AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(-2 - 4)^2 + (2 - 6)^2}$ $= \sqrt{36 + 16}$ $= \sqrt{52}$ $(= 2\sqrt{13})$	Scale 10C(0, 4, 8, 10) <i>Low Partial Credit:</i> - Formula with some substitution <i>High Partial Credit:</i> - Formula fully substituted - Incorrect substitution into formula and completed correctly Note: Accept $1\sqrt{52}$ or $\sqrt{52}$ for full marks
(b) (i)	$\text{Mid pt} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ $= \left(\frac{4 - 2}{2}, \frac{6 + 2}{2} \right)$ $= (1, 4)$	Scale 5C(0, 3, 4, 5) <i>Low Partial Credit:</i> - Formula with some substitution <i>High Partial Credit:</i> - Formula fully substituted - Incorrect substitution into formula and completed correctly <i>Full Credit:</i> - Correct answer without supporting work

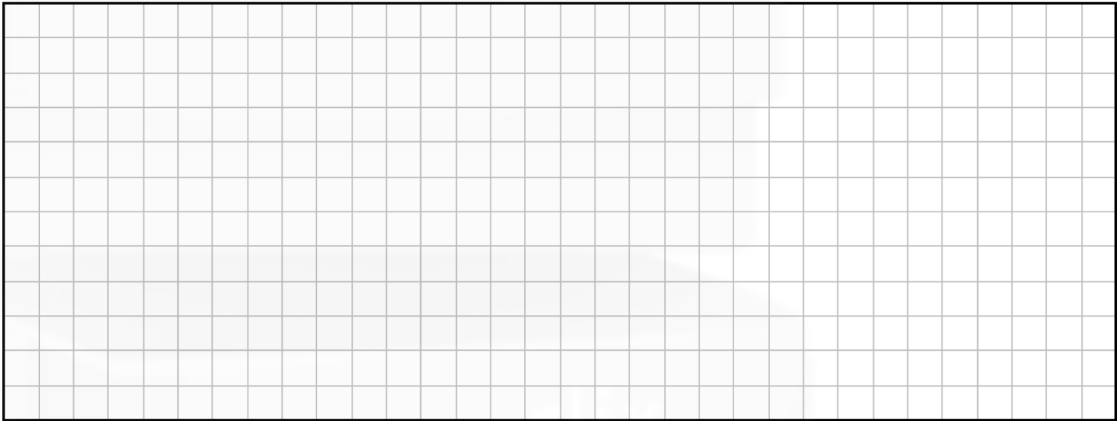
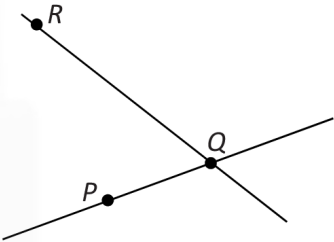
(b) (ii)	$m_{DE} = \frac{3 - 4}{7 - 1} = \frac{-1}{6}$ $m_{BC} = \frac{0 - 2}{10 + 2} = \frac{-2}{12} = -\frac{1}{6}$	Scale 5C(0, 3, 4, 5) <i>Low Partial Credit:</i> • Formula with some substitution • States slopes are equal (or similar) <i>High Partial Credit:</i> • One correct slope • Incorrect substitution into formula and completed correctly
(c)	A(4, 6) B(-2, 2) C(10, 0) x down 10, y no change ↓ (-6, 6) (-12, 2) (0, 0) $\text{Area} = \frac{1}{2} x_1 y_2 - x_2 y_1 $ $= \frac{1}{2} (-6)(2) - (-12)(6) $ $= \frac{1}{2} -12 + 72 $ $= 30 \text{ units}^2$	Scale 5D (0, 2, 3, 4, 5) <i>Low Partial Credit:</i> Work of merit • Translation of any relevant point to (0, 0) indicated • Formula with some substitution <i>Mid Partial Credit:</i> • Formula fully substituted with 2 from A, B, and C <i>High Partial Credit:</i> • Formula fully substituted with translated points • Area of triangle found from formula fully substituted with 2 from A, B, and C • Area of triangle found from an incorrect translation

Question 2

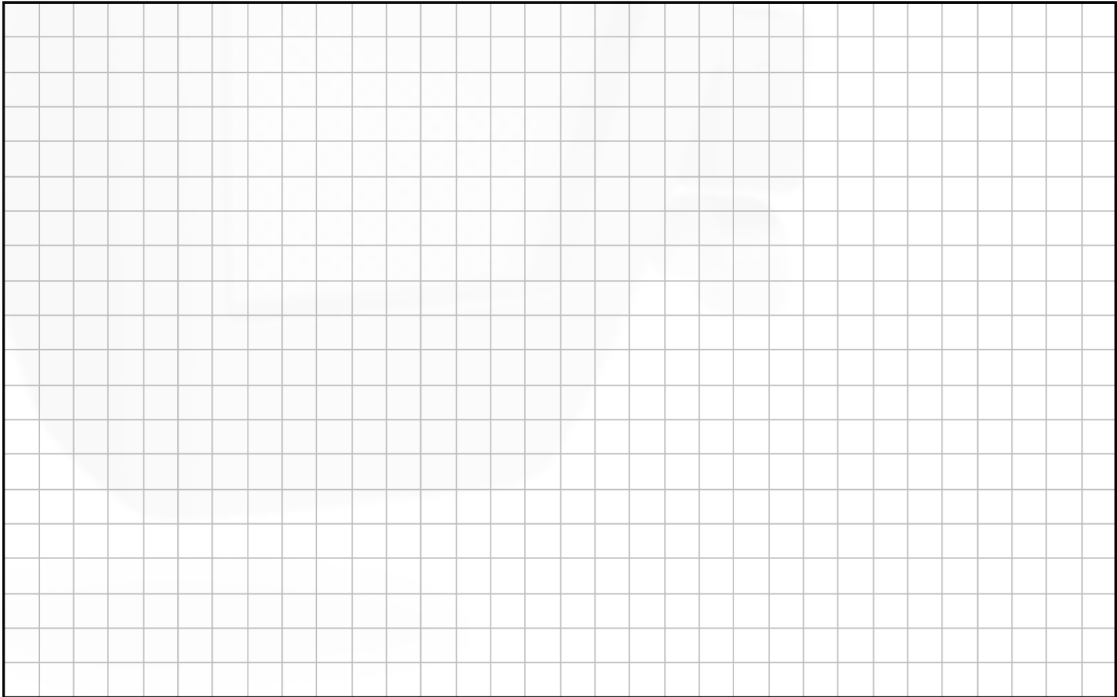
(25 marks)

The diagram shows the line PQ and the line QR .
The co-ordinates of the points are $P(4, 2)$, $Q(8, 5)$ and $R(2, 11)$.

(a) Find the slope of PQ .



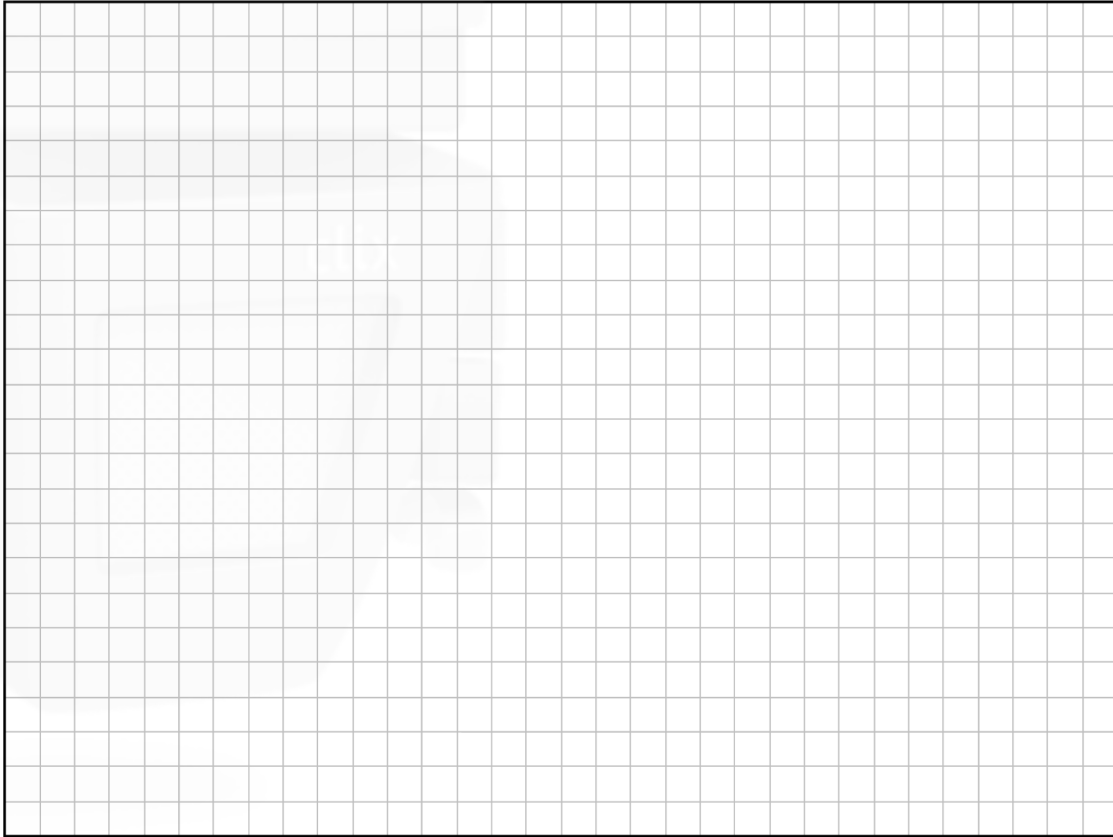
(b) Find the equation of the line PQ .
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.



(c) Write down the slope of any line perpendicular to PQ .

Slope =

(d) Find the area of the triangle PQR .



(a)	$m = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{5 - 2}{8 - 4}$ $= \frac{3}{4}$	Scale 10C(0, 4, 6, 10) <i>Low Partial Credit:</i> • Work of merit • Formula with some correct substitution <i>High Partial Credit:</i> • Formula substituted correctly <i>Full Credit:</i> • Correct answer without work <i>Misreading:</i> • Finds slope of <i>PR</i> or <i>QR</i>
(b)	$y - y_1 = m(x - x_1)$ $y - 2 = \frac{3}{4}(x - 4)$ $3x - 4y - 4 = 0$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Formula with some correct or consistent substitution <i>High Partial Credit:</i> • Formula substituted correctly or consistently • One incorrect substitution followed by correct solution • Answer not in the required format Note: Accept $-3x + 4y + 4 = 0$
(c)	$-\frac{4}{3}$	Scale 5B (0, 2, 5) <i>Mid Partial Credit:</i> • Work of merit <i>Full Credit:</i> • Correct answer without work

(d)

$$\begin{array}{ccc} (4, 2) & (8, 5) & (2, 11) \\ \downarrow & \downarrow & \downarrow \\ (0, 0) & (4, 3) & (-2, 9) \end{array}$$
$$= \frac{1}{2} |x_1 y_2 - x_2 y_1|$$
$$= \frac{1}{2} |(4)(9) - (-2)(3)|$$
$$= \frac{1}{2} |36 + 6|$$
$$= 21 \text{ square units}$$

Scale 5D (0, 2, 3, 4, 5)

Low Partial Credit:

- Translation of any relevant point to (0,0)
- Area of triangle formula with some substitution

Mid Partial Credit:

- Area of triangle formula substituted fully with 2 from P,Q, and R

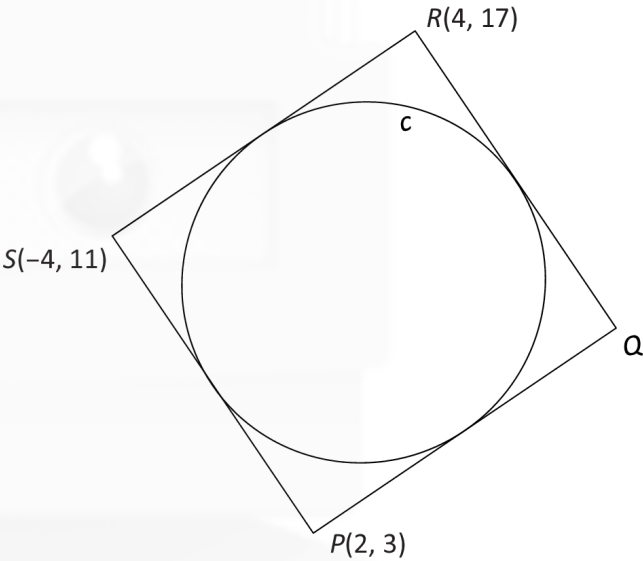
High Partial Credit:

- Formula substituted correctly with translated points
- Area of triangle found from formula substituted fully with 2 from *P, Q, R*
- An incorrect translation applied correctly

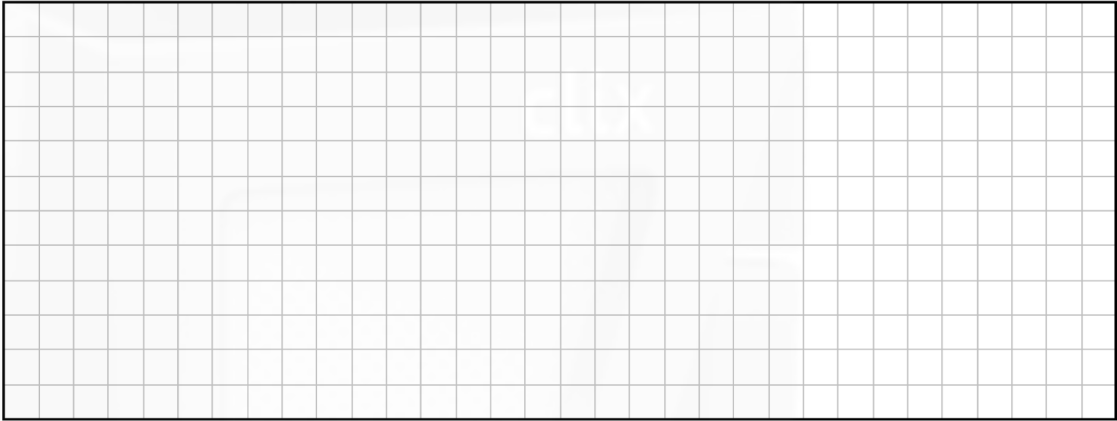
Question 4

(25 marks)

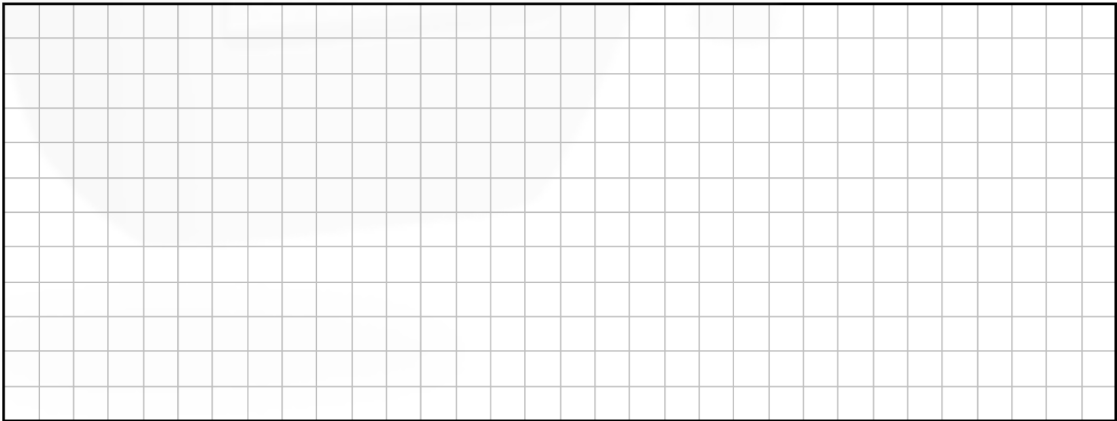
The circle c is enclosed in the square $PQRS$ and touches all four sides, as shown in the diagram. The co-ordinates of three of the vertices are $P(2, 3)$, $R(4, 17)$, and $S(-4, 11)$.



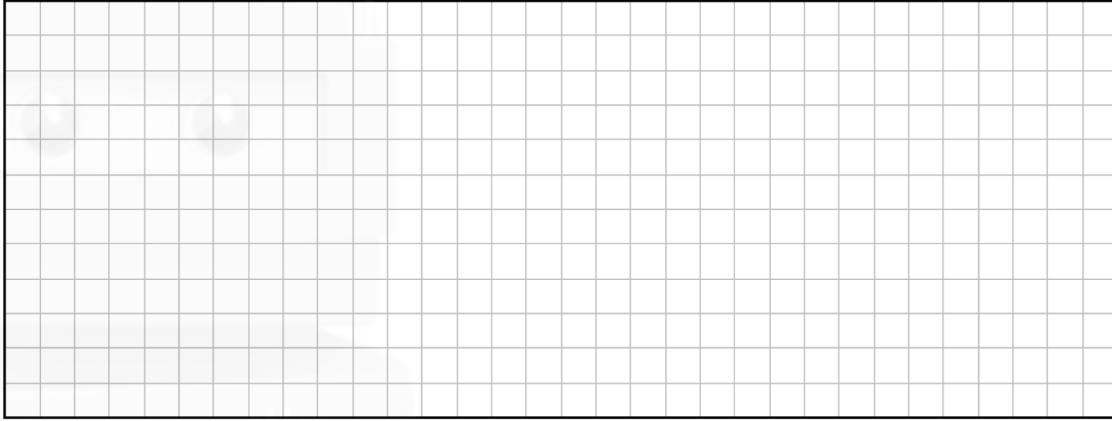
(a) Find the co-ordinates of Q .



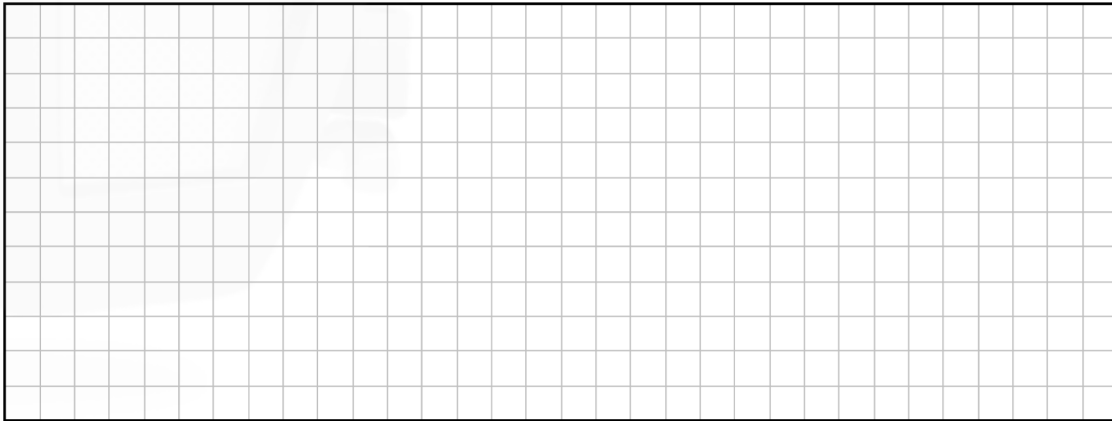
(b) Find the co-ordinates of the centre of c .



(c) Find the length of the **radius** of c .



(d) Find the equation of circle c .



(a)	$S(-4, 11) \rightarrow P(2, 3)$ $\Rightarrow x \uparrow 6 \text{ and } y \downarrow 8$ $R(4, 17) \rightarrow Q(10, 9)$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit • Translation identified or indicated on the diagram <i>High Partial Credit:</i> • One co-ordinate correct • An incorrect translation applied correctly • Correct translation applied to an incorrect point <i>Full Credit:</i> • Correct answer without work
(b)	Mid-point of PR $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ $= \left(\frac{2 + 4}{2}, \frac{3 + 17}{2} \right)$ $= (3, 10)$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Formula with some relevant substitution • Indicates the centre on the diagram <i>High Partial Credit:</i> • Formula substituted correctly <i>Full Credit:</i> • Correct answer without work
(c)	$\text{Radius} = \frac{1}{2} RS \text{ or } \frac{1}{2} SP $ $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(4 + 4)^2 + (17 - 11)^2}$ $= \sqrt{64 + 36} = 10$ $r = \frac{1}{2}(10) = 5 \text{ units}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Distance formula with some relevant substitution <i>High Partial Credit:</i> • Formula substituted correctly • Correct answer without work

(a)

Centre = (3,10) Radius = 5

$$(x - 3)^2 + (y - 10)^2 = 25$$

or

$$x^2 + y^2 - 6x - 20y + 84 = 0$$

Scale 10C (0, 4, 6, 10)

Low Partial Credit:

- Work of merit

High Partial Credit:

- Formula substituted with centre and radius i.e. $(x - 3)^2 + (y - 10)^2 = 5^2$
- One error in substitution and finished correctly

Full Credit:

- Correct or consistent answer without work

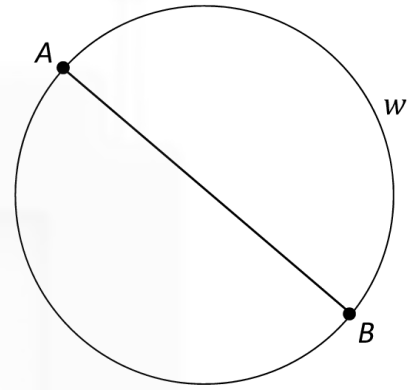
Note:

Accept work with candidates centre and radius

Question 4

(25 marks)

The points $A(1, 8)$ and $B(9, 0)$ are the end-points of a diameter of the circle w , as shown in the diagram.



- (a)** Find the co-ordinates of the centre of w .

[illegible]

- (b)** Find the length of the radius of w . Give your answer in the form $p\sqrt{q}$, where $p, q \in \mathbb{N}$.

- (c)** Hence write down the equation of the circle w .

[illegible]

- (d) Find the equation of the line that is a tangent to the circle w at A .
Give your answer in the form $ax + by + c = 0$, where a, b , and $c \in \mathbb{Z}$.

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

Q4	Model Solution – 25 Marks	Marking Notes
(a)	$\text{Centre} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ $= \left(\frac{1 + 9}{2}, \frac{8 + 0}{2} \right)$ $= (5, 4)$ Method 2 : Translation (1,8) → (Centre) X ↑4, y ↓4 (1,8) → (5,4)	Scale 5B (0, 2, 5) <i>Partial Credit:</i> • Mid-point formula with some correct substitution. • Leaves answer in fraction form. • Indicates centre. • Correct translation identified. <i>Full Credit:</i> • Correct answer without work.
(b)	$ BC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(9 - 5)^2 + (0 - 4)^2}$ $= \sqrt{(4)^2 + (-4)^2}$ $= \sqrt{16 + 16}$ $= \sqrt{32}$ $= 4\sqrt{2} \text{ units}$	Scale 5C(0, 2, 3, 5) <i>Low Partial Credit:</i> • Correct formula with some correct or consistent substitution. <i>High Partial Credit:</i> • Formula fully substituted. • Correct answer without work.
(c)	$(x - h)^2 + (y - k)^2 = r^2$ $(x - 5)^2 + (y - 4)^2 = \sqrt{32}^2$ $(x - 5)^2 + (y - 4)^2 = 32$ Or $x^2 + y^2 - 10x - 8y + 9 = 0$	Scale 10C(0, 3, 7, 10) <i>Low Partial Credit:</i> • Correct formula with some correct or consistent substitution. <i>High Partial Credit:</i> • Answer as $(x - 5)^2 + (y - 4)^2 = \sqrt{32}^2$ <i>Full Credit:</i> • Correct or consistent answer without work.

(d)	$m_r = -1$ $m_T = 1$ $y - 8 = 1(x - 1)$ $x - y + 7 = 0$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Correct relevant formula with some correct substitution. • Finds slope of radius and stops. <i>High Partial Credit:</i> • Eq. of line formula fully substituted. <i>Full Credit:</i> • Answer as $(-x + y - 7 = 0)$. <i>Misreading:</i> • Equation of tangent through B $(x - y - 9 = 0)$.
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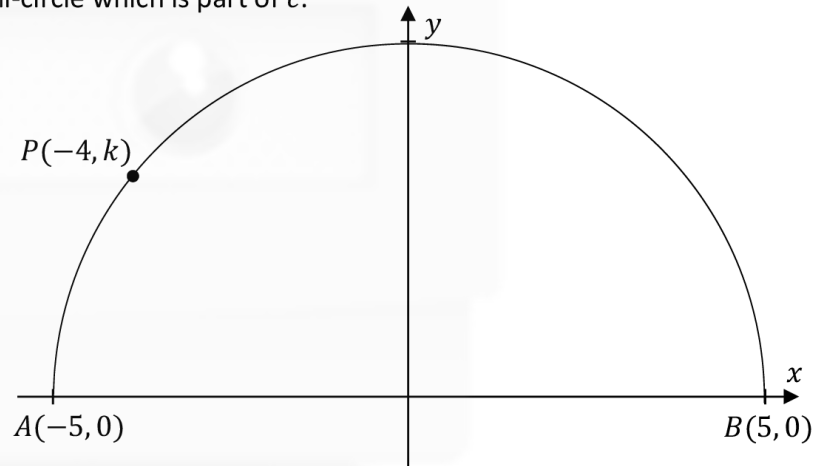
Question 2

(25 marks)

- (a)** The circle c has centre $(0, 0)$ and radius 5 units. Write down the equation of c .

Equation of c : _____

- (b)** The diagram shows a semi-circle which is part of c .

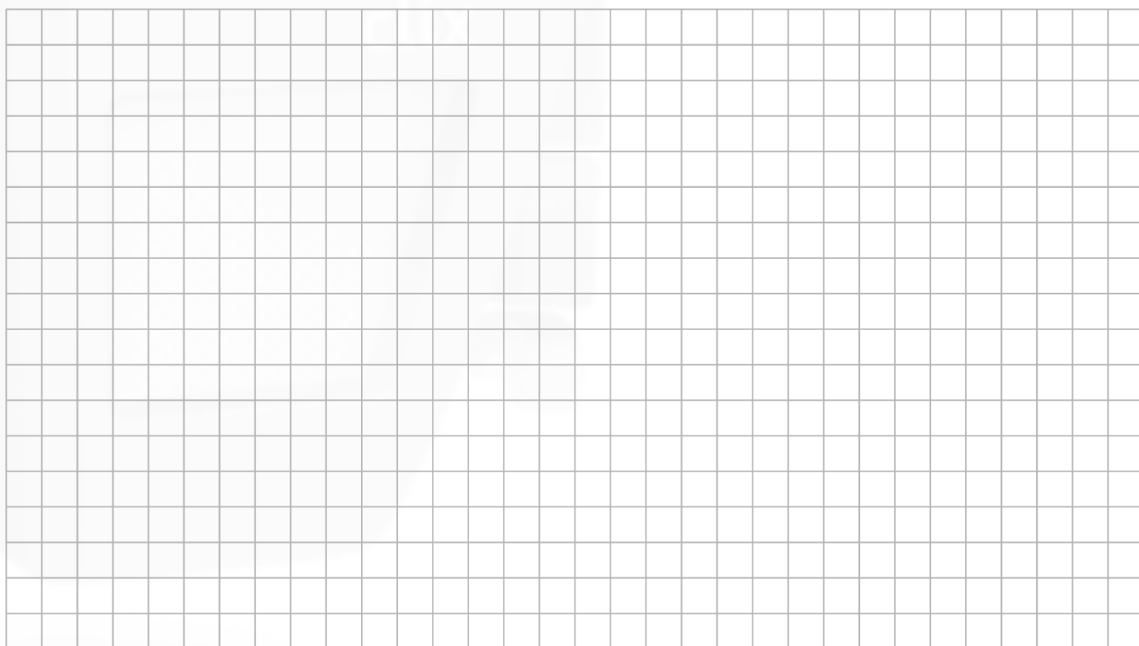
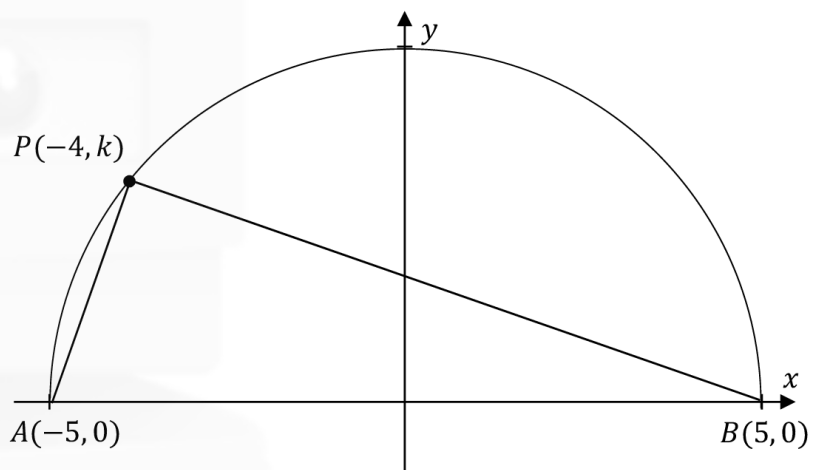


- (i) The point $P(-4, k)$, $k > 0$, is on the semi-circle. Find the value of k .

- (ii)** Show that the triangle ABP is right-angled at P .

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

- (c) Find the area of the region which is inside the semi-circle but outside the triangle ABP .
Give your answer, in square units, correct to 2 decimal places.



(a)	$x^2 + y^2 = 25$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Equation of circle formula with some substitution. e.g. $x^2 + y^2 =$. 25 or 5^2. <i>High Partial Credit:</i> - Equation of circle formula fully substituted. i.e. $x^2 + y^2 = 5^2$
(b) (i)	$16 + k^2 = 25$ $k = 3$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> (-4) or k substituted into equation of circle. <i>High Partial Credit:</i> $(-4)^2 + k^2 = 25$. - Leaves as $k^2 = 9$. - Correct answer without work.
(b) (ii)	AB is the diameter of the circle so the triangle is right-angled at P or $m_{PA} = 3$ $m_{PB} = \frac{-1}{3}$ $m_{PA} \times m_{PB} = -1$ $\Rightarrow PA \perp PB$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Correct relevant formula with some substitution. AB is the diameter (but fails to draw conclusion). - Demonstrates understanding of a requirement to use slopes. e.g. $[m_1 \times m_2 = -1]$ - One correct slope, or length. <i>High Partial Credit:</i> - Both slopes correct but no conclusion. - Triangle ABP is right angled. <i>Zero Credit:</i> - Triangle drawn with P identified as right angled.

(c)

$$A_{\text{region}} = A_{\text{semi-circle}} - A_{\text{triangle}}$$

$$A_{\text{region}} = \frac{1}{2}\pi 5^2 - \frac{1}{2}10(3)$$

$$A_{\text{region}} = 39.2699 - 15$$

$$A_{\text{region}} = 24.2699$$

$$A_{\text{region}} = 24.27$$

Scale 10C (0, 3, 7, 10)

Low Partial Credit:

- Area of circle or semi-circle formula with some substitution.
- Area of triangle formula with some substitution.
- Area = Area of semicircle – Area of triangle (or equivalent).

High Partial Credit:

- Both areas found.
- Answer as $A = \frac{1}{2}\pi 5^2 - \frac{1}{2}10(3)$.

Note:

$$\pi_{\text{Cal}} = 24.27$$

$$\pi_{3.14} = 24.25$$

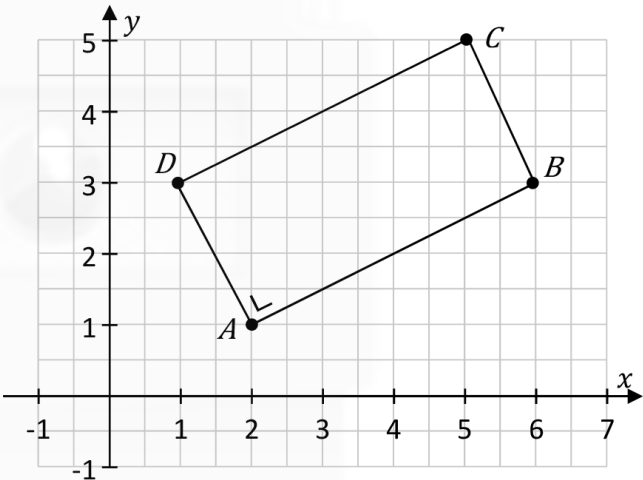
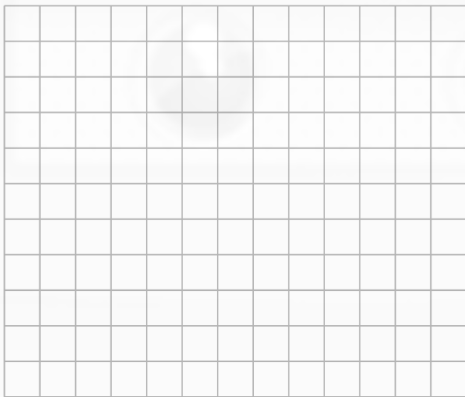
$$\pi_{\frac{22}{7}} = 24.29$$

Question 3

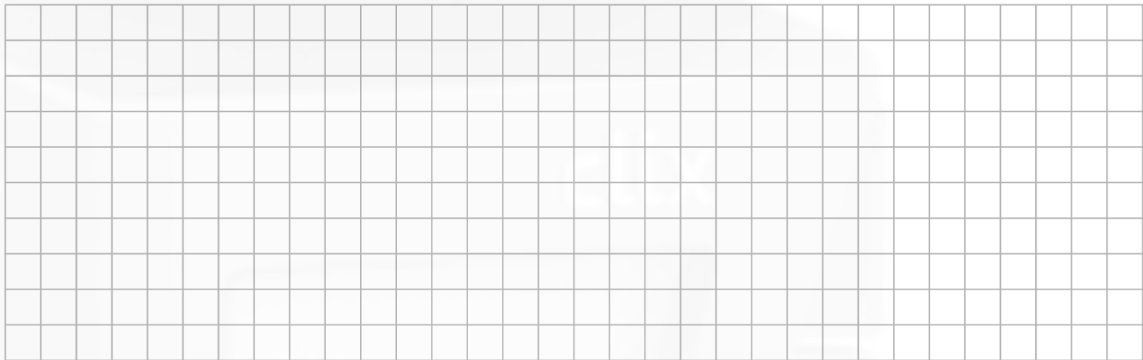
(25 marks)

(a) The points $A(2, 1)$, $B(6, 3)$, $C(5, 5)$, and $D(1, 3)$ are the vertices of the rectangle $ABCD$ as shown.

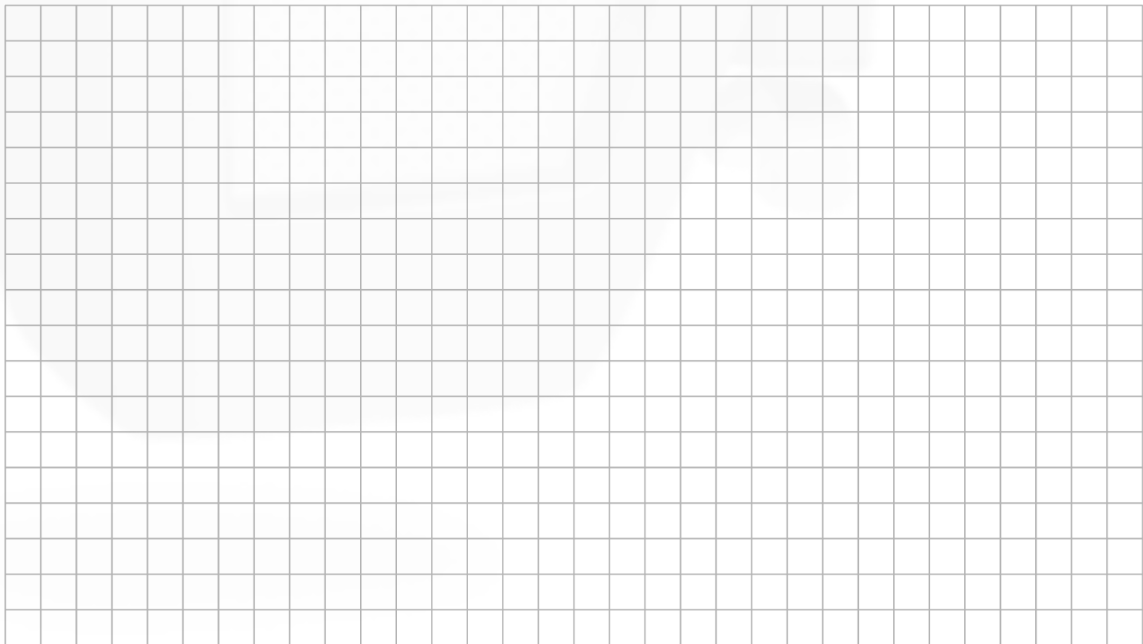
(i) Show that $|AD| = \sqrt{5}$ units.



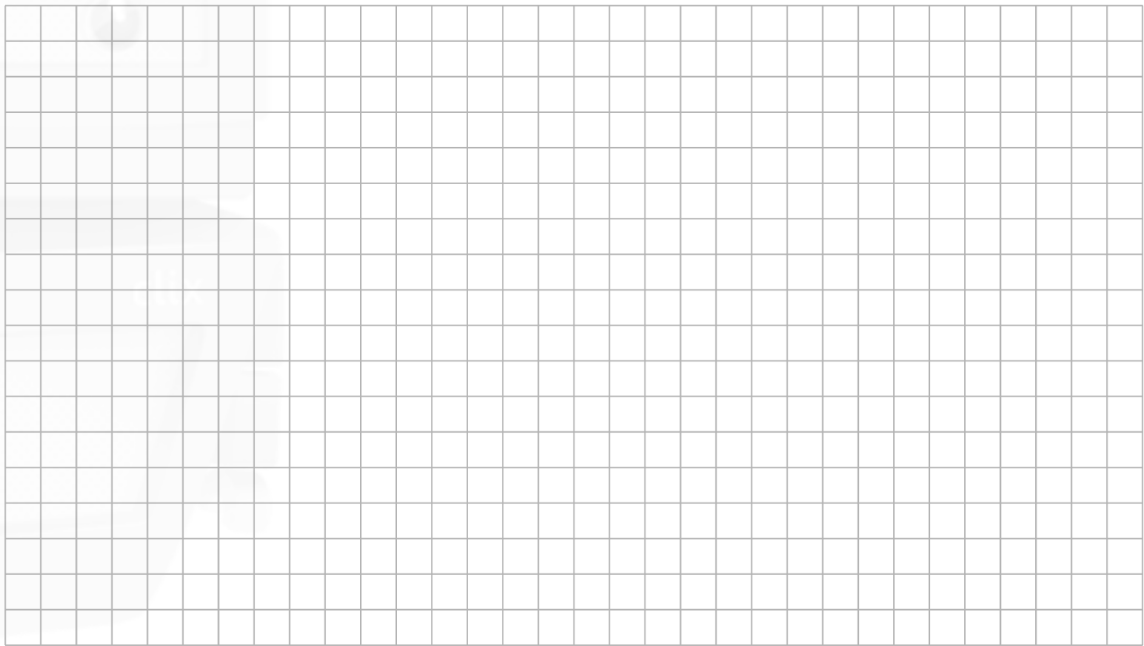
(ii) Find, in square units, the area of the rectangle $ABCD$.

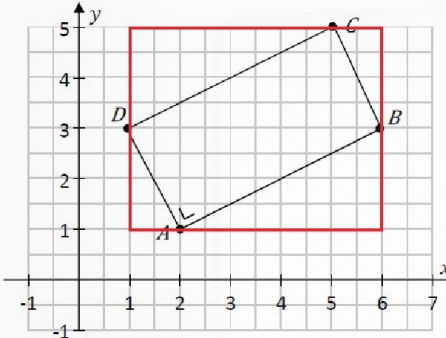


(b) Find the equation of the line BC .
Give your answer in the form $ax + by + c = 0$, where a, b , and $c \in \mathbb{Z}$.



- (c) Use trigonometry to find the measure of the angle ABD .
Give your answer correct to the nearest degree.



(a) (i)	$ AD = \sqrt{1 + 4} = \sqrt{5}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Correct relevant formula with some substitution. • Work with Pythagoras (check candidates diagram) <i>High Partial Credit:</i> • Distance formula fully substituted. • Pythagoras's fully substituted. <i>Zero Credit:</i> • Measurement from diagram (2·2).
(a) (ii)	$ CD = \sqrt{16 + 4} = \sqrt{20}$ $\text{Area} = \sqrt{20} \times \sqrt{5}$ $\text{Area} = 10$ or 	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • $\text{Area} = AD \times CD$. • $\text{Area} = A_{\text{Rectangle}} - 4 A_{\text{Triangles}}$ • Correct relevant formula with some substitution. <i>High Partial Credit:</i> • Area of triangles calculated. • $\text{Area} = \sqrt{20} \times \sqrt{5}$. • $\text{Area} = 20 - (4 + 4 + 1 + 1)$. <i>Zero Credit:</i> • Measurement from diagram (2·2, 4·5). <i>Full Credit:</i> • Correct answer without work.
(b)	$m_{BC} = -2$ $y - 3 = -2(x - 6)$ $2x + y - 15 = 0$	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> • Correct relevant formula with some substitution. <i>High Partial Credit:</i> • Eq. of line formula fully substituted. • One incorrect substitution followed by correct solution. Note: Accept $(-2x - y + 15 = 0)$

(c)

$$\sin ABD = \frac{\sqrt{5}}{5}$$

or

$$\cos ABD = \frac{\sqrt{20}}{5}$$

or

$$\tan ABD = \frac{\sqrt{5}}{\sqrt{20}}$$

$$\text{Showing } \angle ABD = 26.565 \\ = 27^\circ$$

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

- Any work of merit e.g. identifies $\angle ABD$.
- Work of merit with correct trigonometric ratio.
- Correct relevant formula with some substitution.

High Partial Credit:

- Sets up a correct trigonometric ratio.
- Correct relevant formula fully substituted.

Misread (-1):

- Finds $|\angle ADB|$ correctly.

Note:

$$\text{Rad}=0.4636$$

$$\text{Grad}=29.5167$$

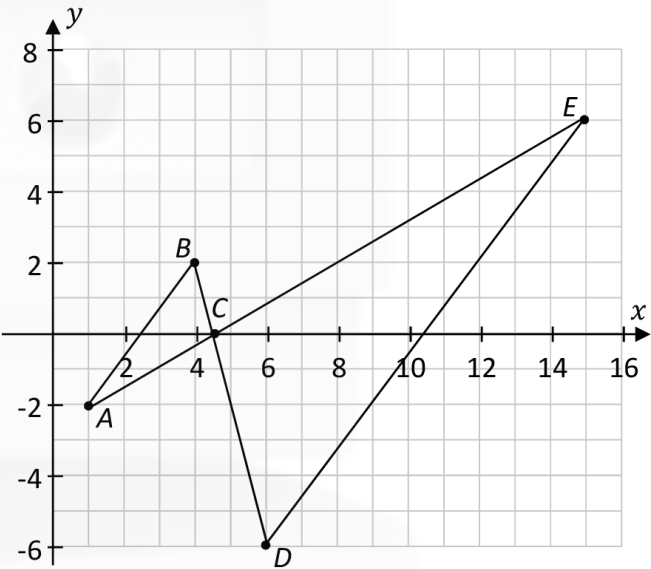
Question 8

(40 marks)

The diagram below shows the triangles CBA and CDE .

- (a) The coordinates of C are $(4.5, 0)$.
From the diagram, write down the coordinates of the points A , B , D , and E .

$A = (\quad , \quad)$
 $B = (\quad , \quad)$
 $D = (\quad , \quad)$
 $E = (\quad , \quad)$



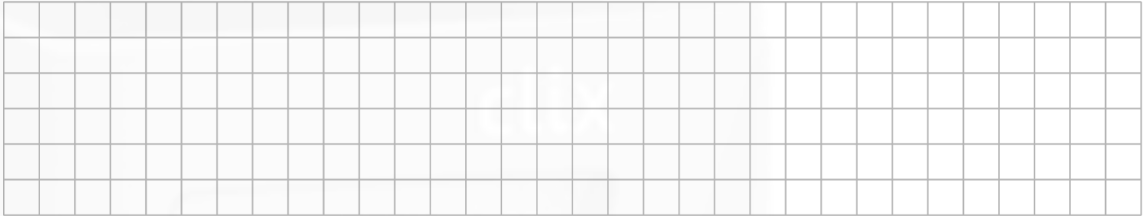
- (b) Show, using slopes, that the line segments $[AB]$ and $[DE]$ are parallel.



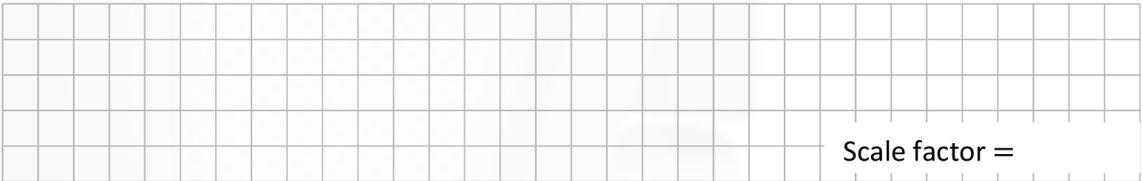
(c) (i) Show that the area of the triangle CBA is 4 square units.



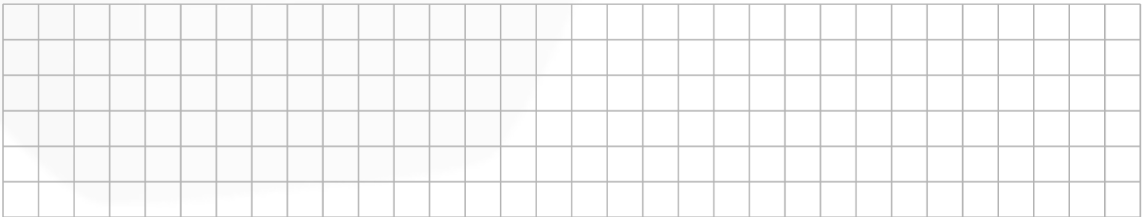
(ii) Find $|AB|$, the distance from A to B .



(iii) The triangle CDE is an enlargement of the triangle CBA .
Given that $|DE| = 15$ units, find the scale factor of the enlargement.



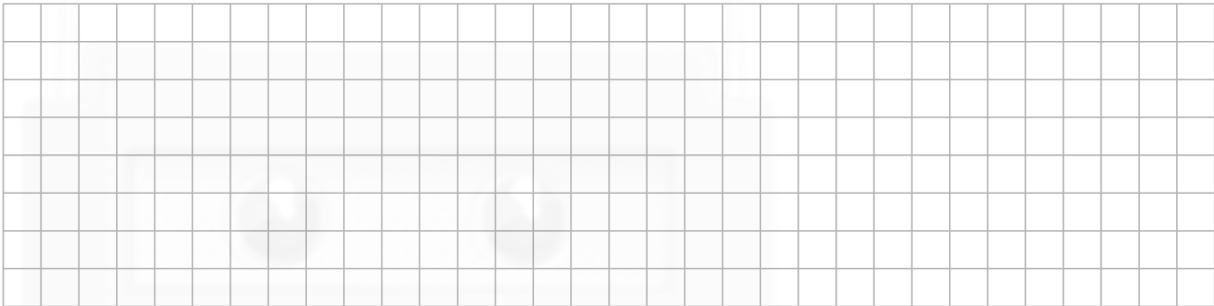
(iv) Use this scale factor to find the area of the triangle CDE .



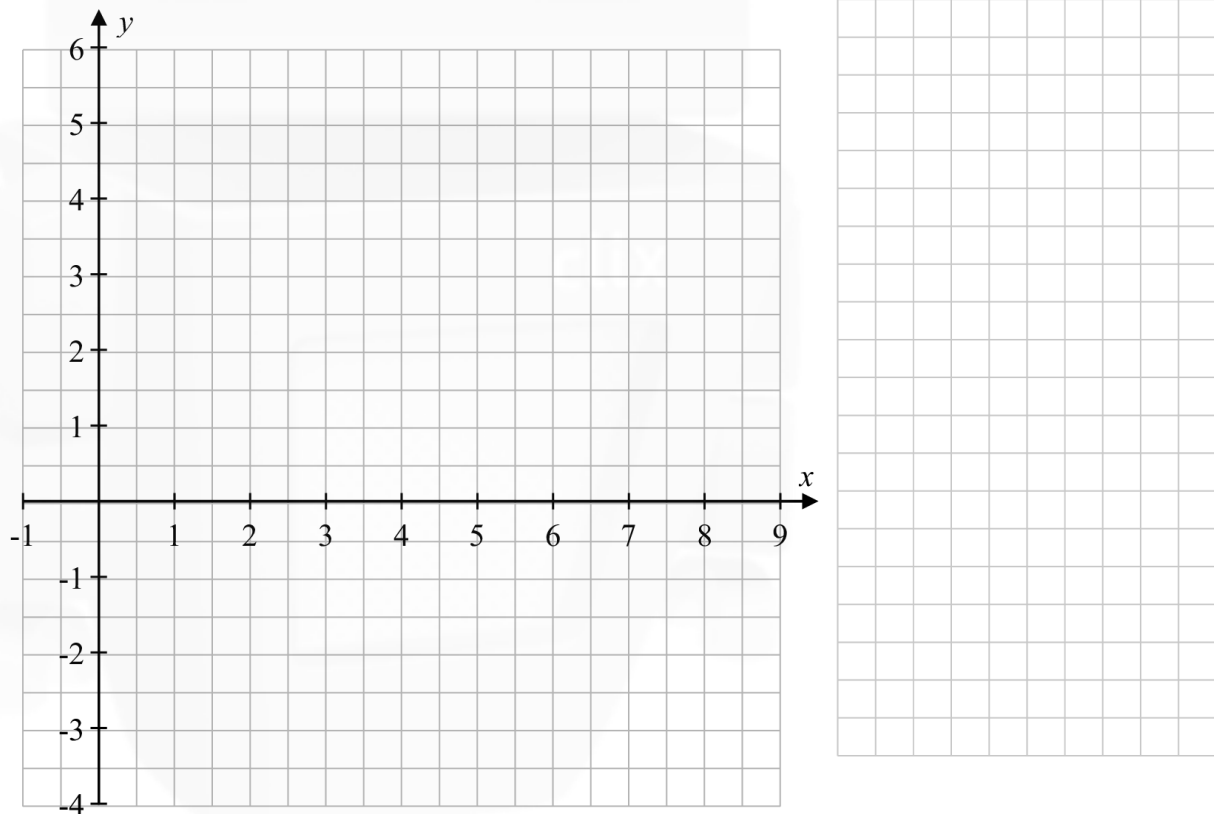
(a)	$A = (1, -2) \quad B = (4, 2)$ $D = (6, -6) \quad E = (15, 6)$	Scale 10C (0, 3, 7, 10) <i>Low Partial Credit:</i> • <u>ONE</u> correct point. <i>High Partial Credit:</i> • <u>THREE</u> correct points. • All answers written consistently as (y, x)
(b)	$m_{AB} = \frac{4}{3}$ $m_{DE} = \frac{12}{9} \text{ or } \frac{4}{3}$ Conclusion: Parallel, or slopes are the same.	Scale 10D (0, 4, 5, 7, 10) <i>Low Partial Credit:</i> • Any mention of same slopes or similar. • Slope formula written with some relevant substitution. • Rise over Run (Written or indicated). <i>Mid Partial Credit:</i> • Slope formula fully substituted for one slope. • One correct slope. <i>High Partial Credit:</i> • Both slopes but no conclusion. • One error in substitution followed by correct calculation and consistent conclusion. <i>Full credit:</i> • Equal slopes and conclusion.
(c) (i)	$A = (1, -2) \quad B = (4, 2) \quad C = (4.5, 0)$ $\downarrow \quad \downarrow \quad \downarrow$ $A = (0, 0) \quad B = (3, 4) \quad C = (3.5, 2)$ $\text{Area} = \frac{1}{2} [(3)(2) - (3.5)(4)]$ $\text{Area} = \frac{1}{2} (6) - (14) $ $= \frac{1}{2} (-8) $ $= 4$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Correct relevant formula with some correct/consistent substitution. • One point translated to (0, 0). <i>High Partial Credit:</i> • $\text{Area} = \frac{1}{2} x_1y_2 - x_2y_1$ fully substituted with translated points. • Error in translation and area calculated correctly. • No translation but uses TWO of the points A, B, C and area calculated correctly. • One error in substitution followed by correct calculation.

(c) (ii)	$ AB = \sqrt{3^2 + 4^2} = 5 \text{ units}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Correct relevant formula with some substitution. • Any work of merit indicated in this part. <i>High Partial Credit:</i> • Correct substitution without calculation. • One error in substitution followed by correct calculation. • Correct answer without work.
(c) (iii)	$\frac{15}{5} = 3 = k$	Scale 5B (0, 3, 5) <i>Partial Credit:</i> • Use of answer to part (c)(ii). • Scale factor formula written or indicated. • Attempts to calculate AB in this part. <i>Full Credit:</i> • Correct answer without work.
(c) (iv)	$\text{Area} = 4 \times 3^2 = 36 \text{ units}^2$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work of merit with answer from (c)(iii). • Work of merit with answer from (c)(i). <i>High Partial Credit:</i> • Area as $3 \times 4 = 12$ or equivalent. <i>Full Credit:</i> • Correct answer without work.

- (a) The line l contains the points $A(4, 5)$ and $B(2, 0)$. Find the equation of l .
Give your answer in the form $ax + by + c = 0$ where a, b , and $c \in \mathbb{Z}$.

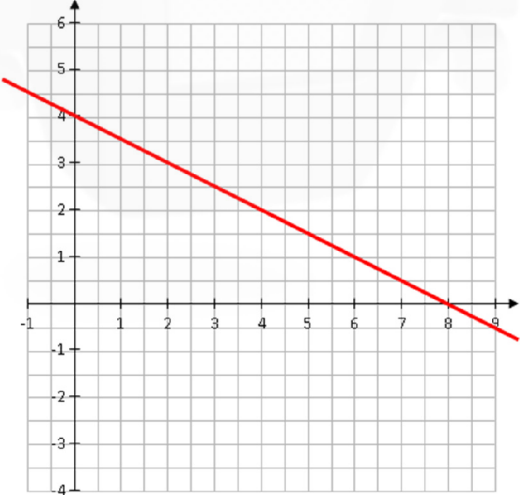


- (b) Draw the line $k: x + 2y = 8$ on the axes below.



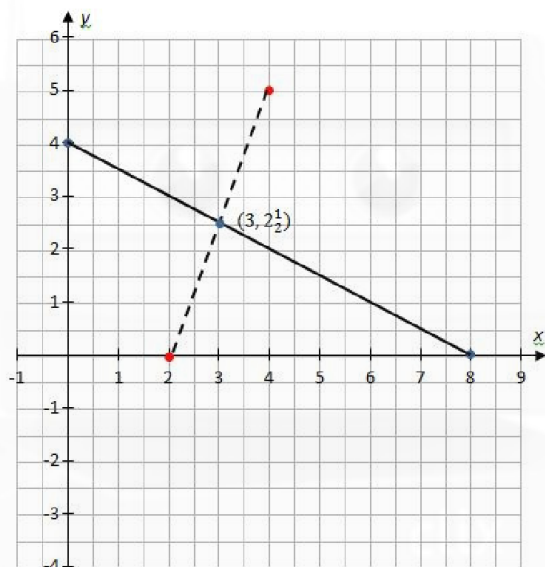
- (c) Use a graphic, numeric or algebraic method to find the co-ordinates of $l \cap k$.



Q4	Model Solution – 25 Marks	Marking Notes
(a)	$m = \frac{0 - 5}{2 - 4} = \frac{5}{2}$ $y - 5 = \frac{5}{2}(x - 4)$ $2y - 10 = 5x - 20$ $5x - 2y - 10 = 0$	Scale 15D(0, 2, 4, 8, 15) <i>Low Partial Credit:</i> - Identifies a correct relevant formula. <i>Mid Partial Credit:</i> - Slope formula with some substitution. - Equation of line formula with some substitution. - Slope correct and stops. <i>High Partial Credit:</i> - Equation of line formula fully substituted correctly (consistently). Note: Accept $(-5x + 2y + 10 = 0, \text{ in any order})$.
(b)	 $x = 0, y = 4 \Rightarrow (0, 4)$ $y = 0, x = 8 \Rightarrow (8, 0)$	Scale 5C(0, 1, 2, 5) <i>Low Partial Credit:</i> $x = 0$ or $y = 0$ written. - Any work of merit working with line k. <i>High Partial Credit:</i> - Finds two correct points but no plots or line on graph.

(c)

Graphical:



Numerical:

$$\begin{aligned}5x - 2y - 10 &= 0 \\5(3) - 2y - 10 &= 0 \\15 - 2y - 10 &= 0 \\2y &= 5 \\y &= 2\frac{1}{2} \\x + 2y &= 8 \\3 + 2y &= 8 \\2y &= 5 \\y &= 2\frac{1}{2} \\l \cap k &= (3, 2\frac{1}{2})\end{aligned}$$

Algebraic:

$$\begin{aligned}x + 2y &= 8 \\5x - 2y &= 10 \\6x &= 18 \\x &= 3 \\y &= \frac{5}{2} \\&\left(3, 2\frac{1}{2}\right)\end{aligned}$$

Scale 5C(0, 1, 2, 5)

Low Partial Credit:

- Effort to draw line l .

High Partial Credit:

- l drawn correctly on grid (in this part) and point of intersection marked but not written.

Full Credit:

- $l \cap k = (3, 2\frac{1}{2})$.

Low Partial Credit:

- Effort at substituting a value of x into equations of one or both lines.

High Partial Credit:

- Uses $x = 3$ to find $y = 2\frac{1}{2}$ in both equations.

Full Credit:

- $l \cap k = (3, 2\frac{1}{2})$.

Low Partial Credit:

- Effort to solve using simultaneous equations.

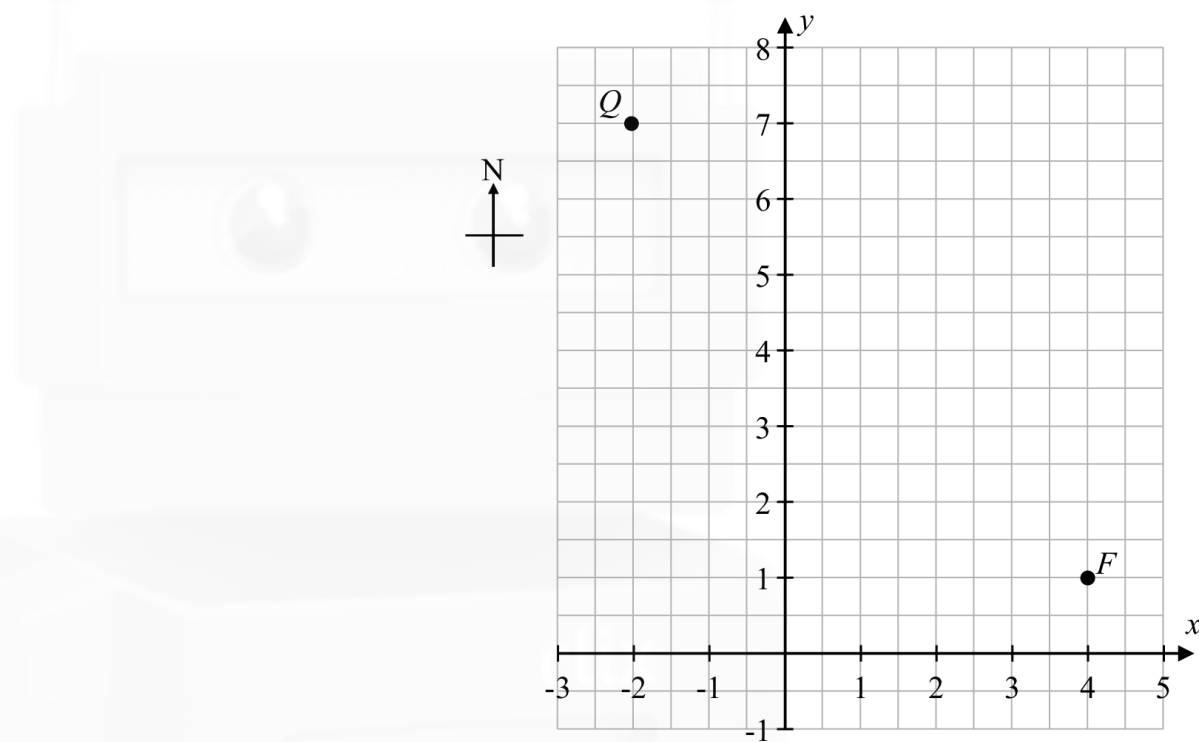
High Partial Credit:

- x and/ or y calculated correctly.
- Writes $l \cap k$ as (y, x) .

Full Credit:

- $l \cap k = (3, 2\frac{1}{2})$.

Joe wants to draw a diagram of his farm. He uses axes and co-ordinates to plot his farmhouse at the point F on the diagram below.



- (a) (i) Write down the co-ordinates of the point F .

$F = (\quad , \quad)$

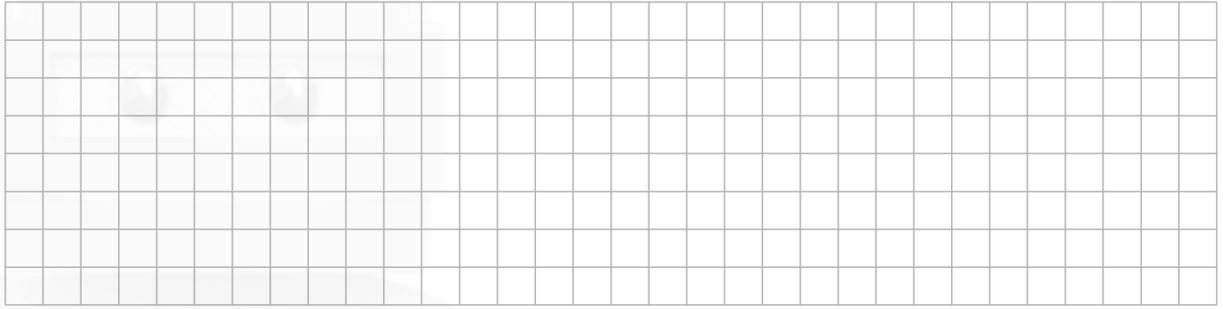
- (ii) A barn is 5 units directly North of the farmhouse. Plot the point representing the position of the barn on the diagram. Label this point B .
- (b) Joe's quad bike is marked with the point Q on the diagram. Find the distance from the barn (B) to the quad (Q). Give your answer correct to 2 decimal places.



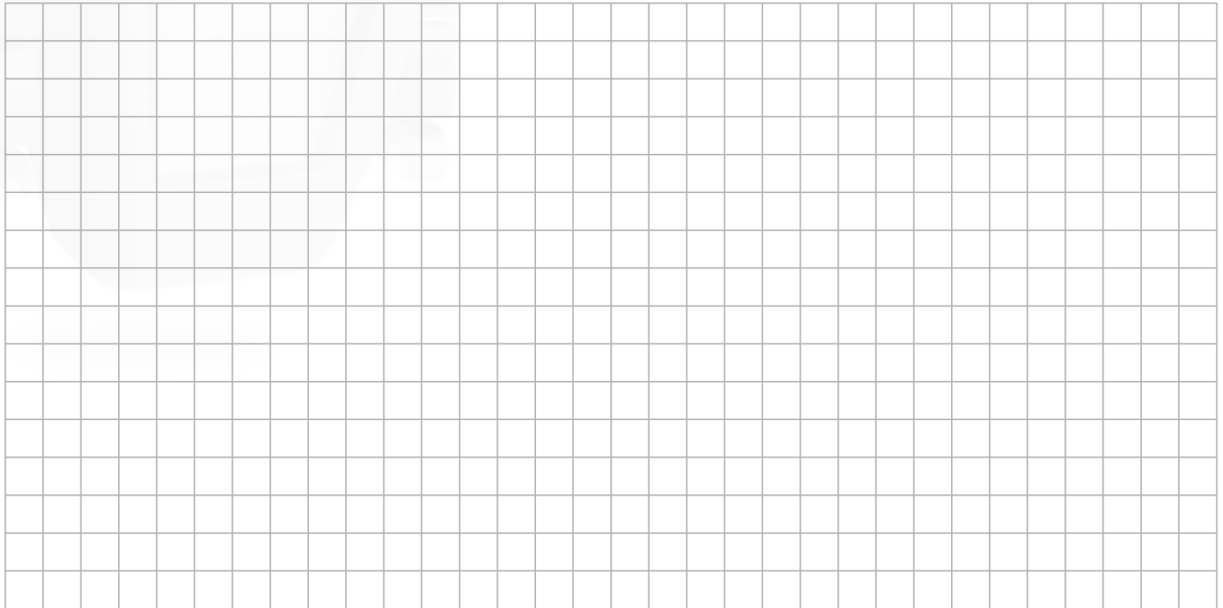
- (c) Joe's tractor is at the point T , where $FBQT$ is a parallelogram. Plot T on the diagram and write the co-ordinates of T in the space below.

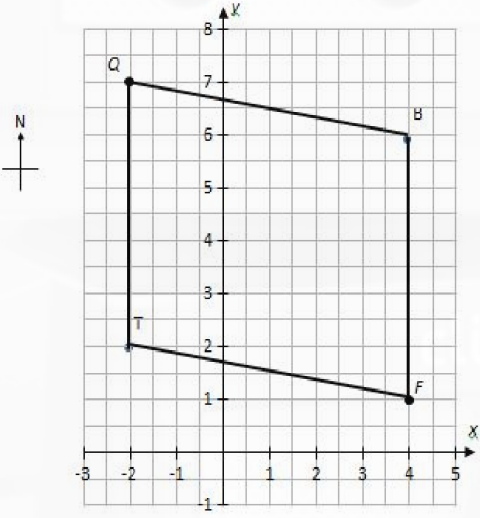
$T = (\quad , \quad)$

- (d) Joe wants to plough the land enclosed by the parallelogram $FBQT$.
Find the area of this parallelogram in square units.



- (e) Given that $|\angle QFB| = 45^\circ$, use trigonometric methods to find $|\angle BQF|$.
Give your answer in degrees correct to one decimal place.



Q9	Model Solution – 45 Marks	Marking Notes
(a) (i)	$F = (4, 1)$	Scale 5B(0, 1, 5) <i>Partial Credit:</i> • One ordinate correct. • Answer as (1, 4).
(a) (ii)		Scale 5B(0, 1, 5) <i>Partial Credit:</i> • Point correct. (No label) • Point plotted as (B), but not in the correct position. • Plot and label as (4, 6). • Any plot North of F. <i>No Credit:</i> • All other incorrect points with no labels.
(b)	$d = \sqrt{(-2 - 4)^2 + (7 - 6)^2}$ $= \sqrt{37}$ $= 6.08$	Scale 5C(0, 1, 2, 5) <i>Low Partial Credit:</i> • Writes co-ordinates of B and/or Q. • Any work of merit. <i>High Partial Credit:</i> • Fully correct/consistent substitution without calculation. • One incorrect substitution with correct calculation. • One consistent error throughout.
(c)	$T = (-2, 2)$	Scale 5B(0, 1, 5) <i>Partial Credit:</i> • Any work of merit. (Join F to B) • One ordinate correct. • Answer as (2, -2).

(d)	$\begin{array}{ccc} (-2, 2), & (4, 6), & (4, 1) \\ & \downarrow & \\ (0, 0), & (6, 4), & (6, -1) \end{array}$ $\text{Area} = 2 \times \text{Area } \triangle TBF$ $2 \times \left(\frac{1}{2}\right) x_1 y_2 - x_2 y_1 $ $ (6)(-1) - (4)(6) $ $= 30$ or $\text{Area} = a \times h_{\perp} = 5 \times 6 = 30$	Scale 5D(0, 2, 3, 4, 5) <i>Low Partial Credit:</i> • Correct relevant formula identified. • Triangle co-ordinates identified correctly. • Triangle translated correctly. • Works with 5, 6, or 6.08. <i>Mid Partial Credit:</i> • Triangle formula fully substituted. • One error in substitution followed by correct calculation of triangle. <i>High Partial Credit:</i> • Area of triangle calculated correctly. • One error in substitution followed by correct calculation of parallelogram. • Parallelogram formula fully substituted. • A correct answer without work.
(e)	$\frac{\sin \angle FQB}{ FB } = \frac{\sin 45^\circ}{ QB }$ $\sin \angle FQB = \frac{5 \times \sin 45^\circ}{6.08}$ $\sin \angle FQB = 0.5815$ $ \angle FQB = 35.6^\circ$	Scale 20C(0, 5, 10, 20) <i>Low Partial Credit:</i> • Correct relevant formula. • Work of merit on diagram. <i>High Partial Credit:</i> • Sine rule fully substituted. • $\angle BQF$ calculated with one incorrect substitution. • $\angle QBF = 99.47^\circ$. • Incorrect calculator mode. (once only) Radian = 0.8 Gradian = 35.9 • Correct answer without work. <i>No Credit:</i> • Treats as right-angled or isosceles Triangle.

The function $f: x \mapsto x^3 + x^2 - 2x + 7$ is defined for $x \in \mathbb{R}$.

- (a)** Find the coordinates of the point at which the graph of f cuts the y -axis.

[illegible]

- (b)** Verify, using algebra, that the point $A(1, 7)$ is on the graph of f .

- (c) (i)** Find $f'(x)$, the derivative of $f(x)$.
Hence find the slope of the tangent to the graph of f when $x = 1$.

$f'(x) =$

Slope =

- (ii)** Hence, find the equation of the tangent to the graph of f at the point $A(1, 7)$.

A full-page view of a blank sheet of graph paper. The grid consists of thin, light gray horizontal and vertical lines forming small squares across the entire page. There are no margins, text, or other markings on the paper.

Q4	Model Solution – 25 Marks	Marking Notes
(a)	$x = 0$ $y = 7$	Scale 5B (0, 2, 5) <i>Partial Credit</i> $x = 0$, written - Zero correctly substituted into equation
(b)	$1 + 1 - 2 + 7 = 7$ $7 = 7$	Scale 5B (0, 2, 5) <i>Partial Credit</i> Any correct substitution
(c) (i)	$f'(x) = 3x^2 + 2x - 2$ at $x = 1$ $f'(x) = 3(1)^2 + 2(1) - 2$ $m = 3$	Scale 10D (0, 2 4, 6, 10) <i>Low Partial Credit</i> - Any correct differentiation <i>Mid Partial Credit</i> - Fully correct differentiation <i>High Partial Credit</i> - Correct substitution for slope
(c) (ii)	$y - 7 = 3(x - 1)$ or $y = 3x + 4$ or $3x - y + 4 = 0$	Scale 5B (0, 2, 5) <i>Partial Credit</i> - Shows $m = 3$ - Correct equation of a line formula - Any correct substitution

Question 2

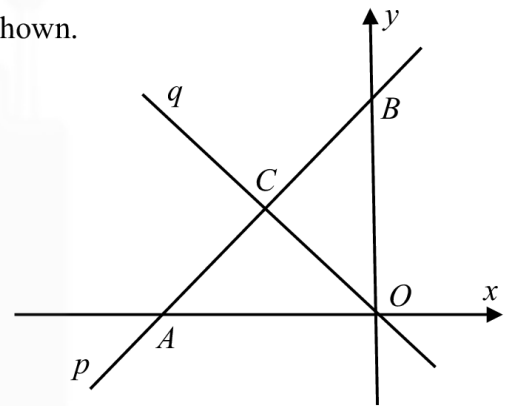
(25 marks)

The line p makes equal intercepts on the axes at A and at B , as shown.

- (a) (i)** Write down the slope of p .

Slope of $p =$ _____

- (ii) The point $(1, 5)$ is on p . Find the equation of p . Write your answer in the form $ax + by + c = 0$, where a, b , and $c \in \mathbb{Z}$.

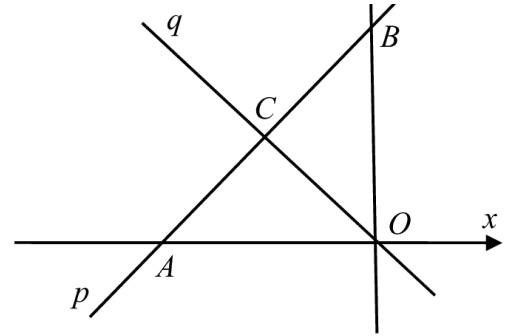


- (b)** The line q is perpendicular to p and contains the point $O(0, 0)$. Find the equation of q .

- (c) The lines p and q intersect at the point C . Explain why the triangles OCA and OBC are congruent.

- (a) (i) Write down the slope of p .

Slope of $p = 1$



- (ii) The point $(1, 5)$ is on p . Find the equation of p .
Write your answer in the form $ax + by + c = 0$,
where a, b , and $c \in \mathbb{Z}$

$$y - y_1 = m(x - x_1)$$

$$y - 5 = 1(x - 1) \Rightarrow x - y + 4 = 0$$

- (b) The line q is perpendicular to p and contains the point $O(0, 0)$. Find the equation of q .

$$q \perp p \Rightarrow \text{slope of } q = -1$$

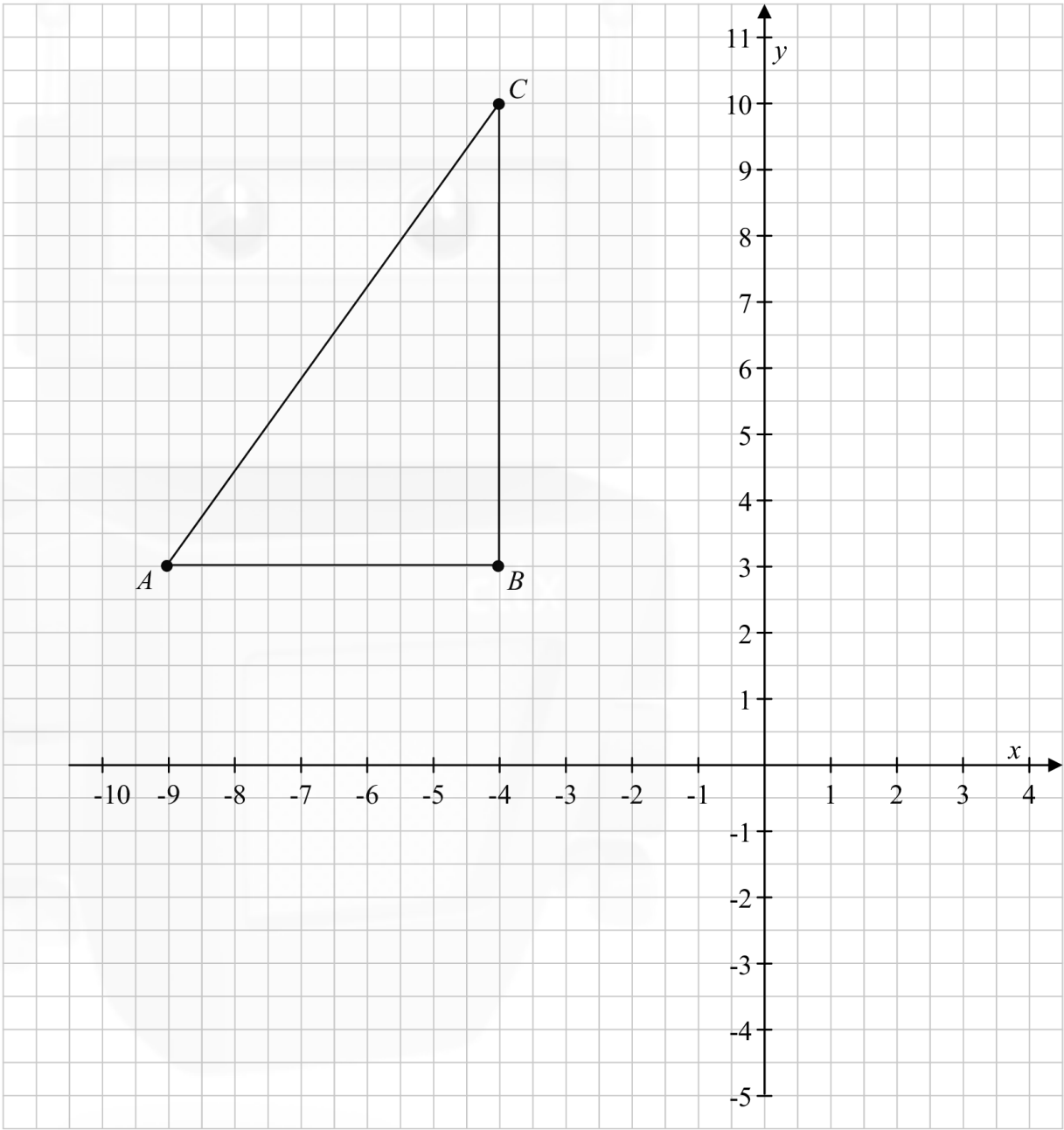
$$y - y_1 = m(x - x_1)$$

$$y - 0 = -1(x - 0) \Rightarrow x + y = 0$$

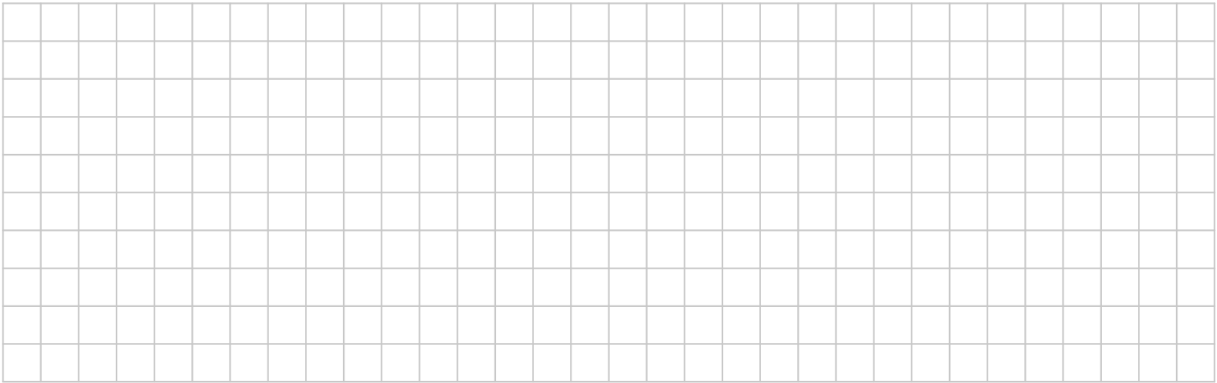
- (c) The lines p and q intersect at the point C . Explain why the triangles OCA and OBC are congruent.

In $\triangle OCA$ and $\triangle OBC$,
 $|OA| = |OB|$... equal intercepts
 $|OC| = |OC|$... common line segment
 $|\angle OCA| = |\angle BCO|$... right angle
 Giving R.H.S.
 Hence, the triangles OCA and OBC are congruent.

The points $A(-9, 3)$, $B(-4, 3)$ and $C(-4, 10)$ are the vertices of the triangle ABC , as shown.



- (a) (i) Find the length of $[AB]$.



(ii) Find the area of the triangle ABC .



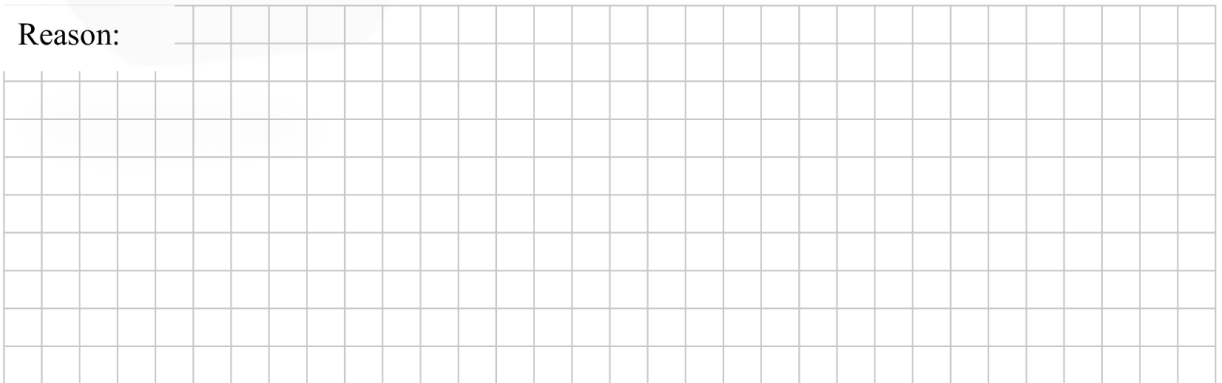
(b) $X(2, -4)$ and $Y(2, 1)$ are two points.

(i) Draw, on the diagram above, a triangle, XYZ , which is congruent to the triangle ABC .

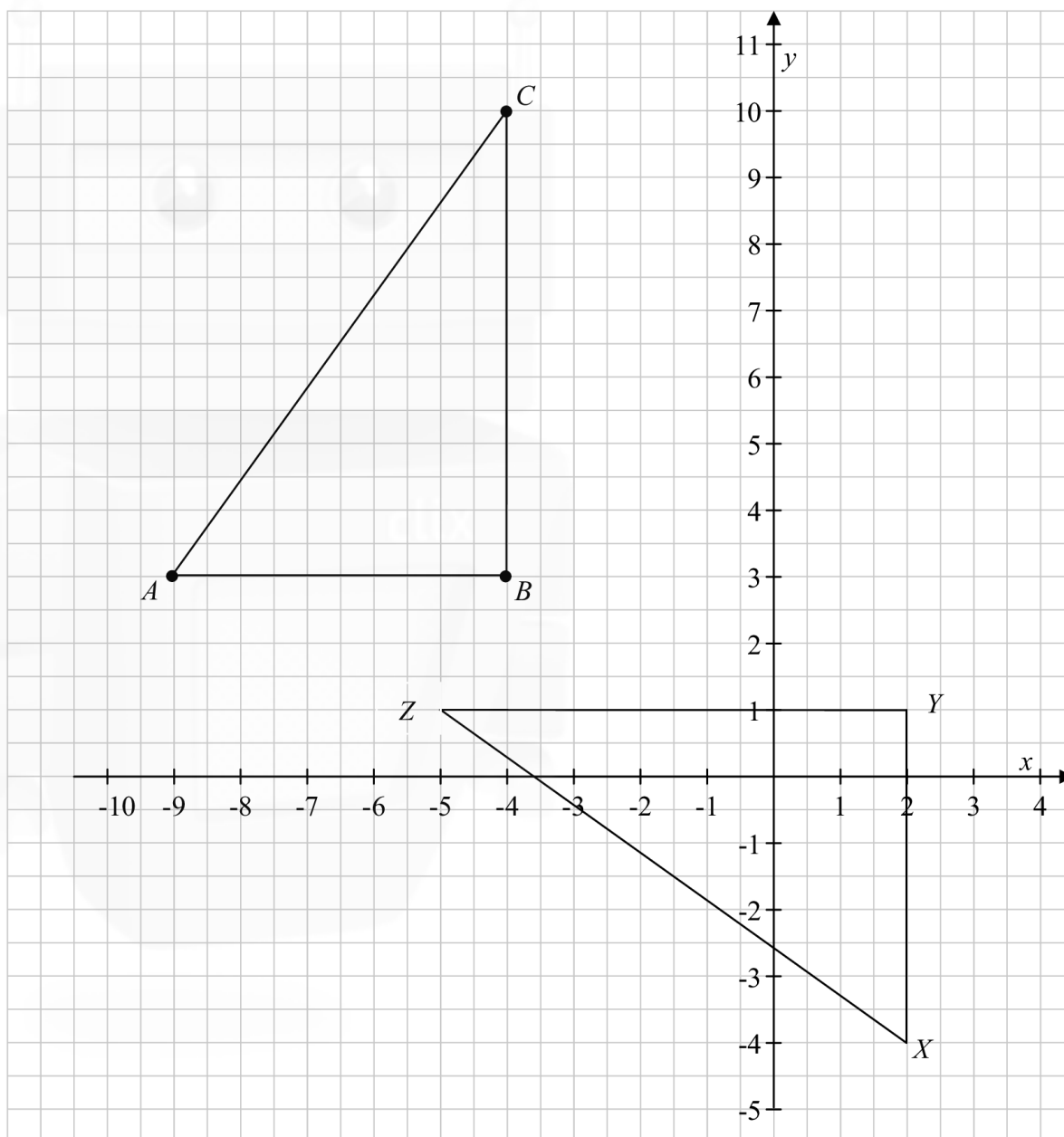
(ii) Write down the co-ordinates of Z and explain why the triangle XYZ is congruent to the triangle ABC .

$Z = (\quad , \quad)$

Reason:



The points $A(-9, 3)$, $B(-4, 3)$ and $C(-4, 10)$ are the vertices of the triangle ABC , as shown.



- (a) (i) Find the length of $[AB]$.

$$\sqrt{(3-3)^2 + (-4+9)^2} = 5$$

or

$$5$$

(ii) Find the area of the triangle ABC .

$$\begin{aligned} \text{Area} &= \frac{1}{2}(5)(7) \\ &= 17\frac{1}{2} \end{aligned}$$

or

$$\begin{aligned} &(-9, 3) \quad (-4, 3) \quad (-4, 10) \\ &x \uparrow 9, \quad y \downarrow 3 \\ &(0, 0) \quad (5, 0) \quad (5, 7) \\ \text{Area} &= \frac{1}{2}(5 \times 7 + 5 - 0) \\ &= 17\frac{1}{2} \end{aligned}$$

(b) $X(2, -4)$ and $Y(2, 1)$ are two points.

(i) Draw, on the diagram above, a triangle, XYZ , which is congruent to the triangle ABC .

(ii) Write down the co-ordinates of Z and explain why the triangle XYZ is congruent to the triangle ABC .

$$Z = (\quad , \quad)$$

$$Z = (-5, 1)$$

or

$$(-5, -4)$$

Reason:

SSS

$$|AB| = |XY| \quad S$$

$$|BC| = |YZ| \quad S$$

$$|AC| = |XZ| \quad S$$

Question 3

(25 marks)

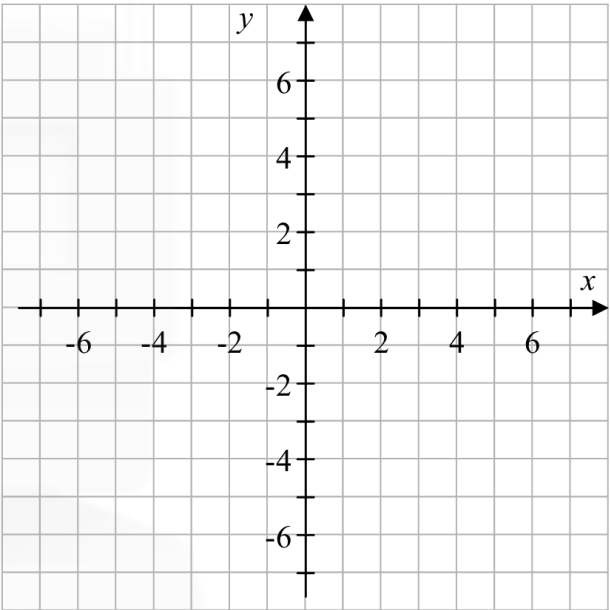
The points A , B , and C have co-ordinates as follows:

$A (3, 5)$

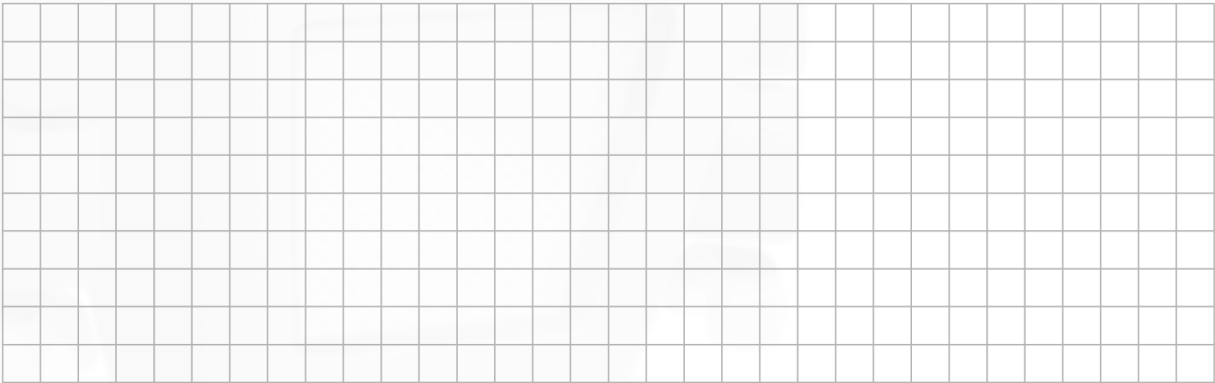
$B (-6, 2)$

$C (4, -4)$.

- (a) Plot A , B , and C on the diagram.



- (b) Find the equation of the line AB .



- (c) Find the area of the triangle ABC .



Question 3

(25 marks)

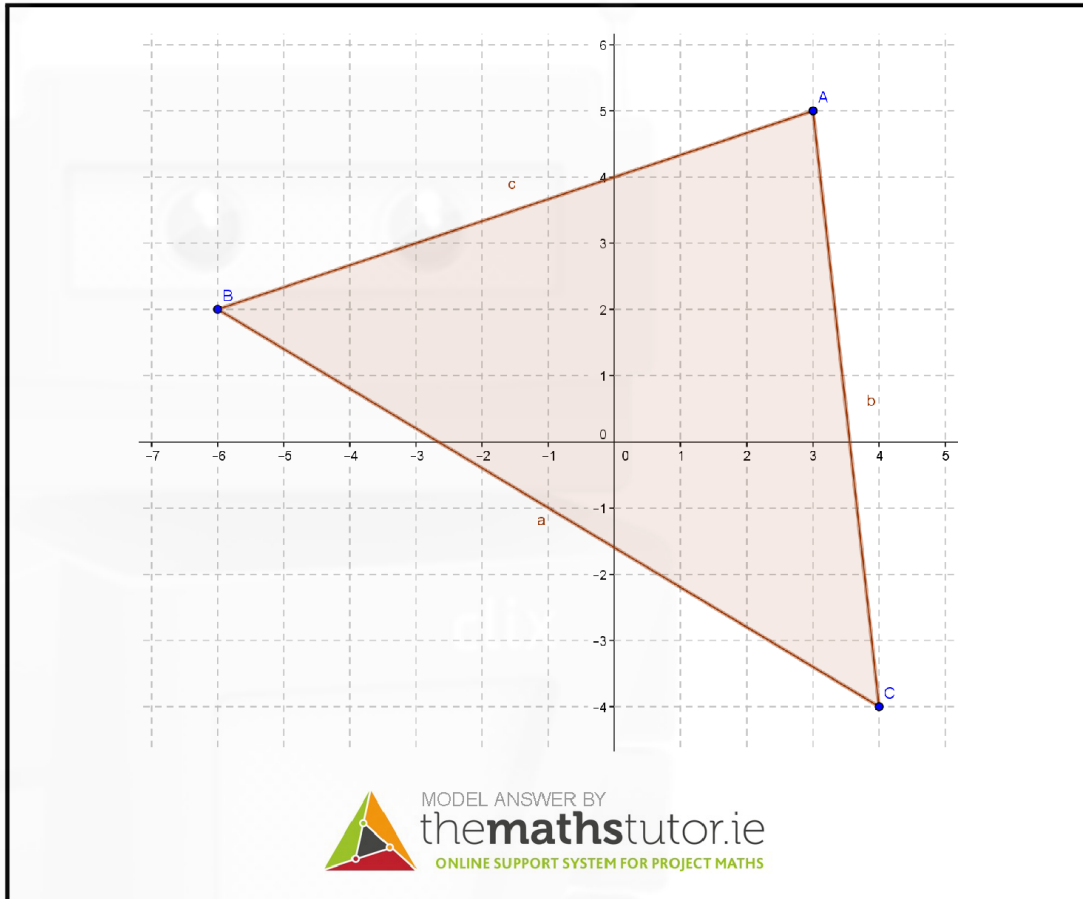
The points A , B , and C have co-ordinates as follows:

$$A(3, 5)$$

$$B(-6, 2)$$

$$C(4, -4)$$

(a) Plot A , B , and C on the diagram.



(b) Find the equation of the line AB .

The slope of the line passing through A and B is

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 2}{3 - (-6)} = \frac{3}{9} = \frac{1}{3}$$

Thus, the line passing through A and B is given by

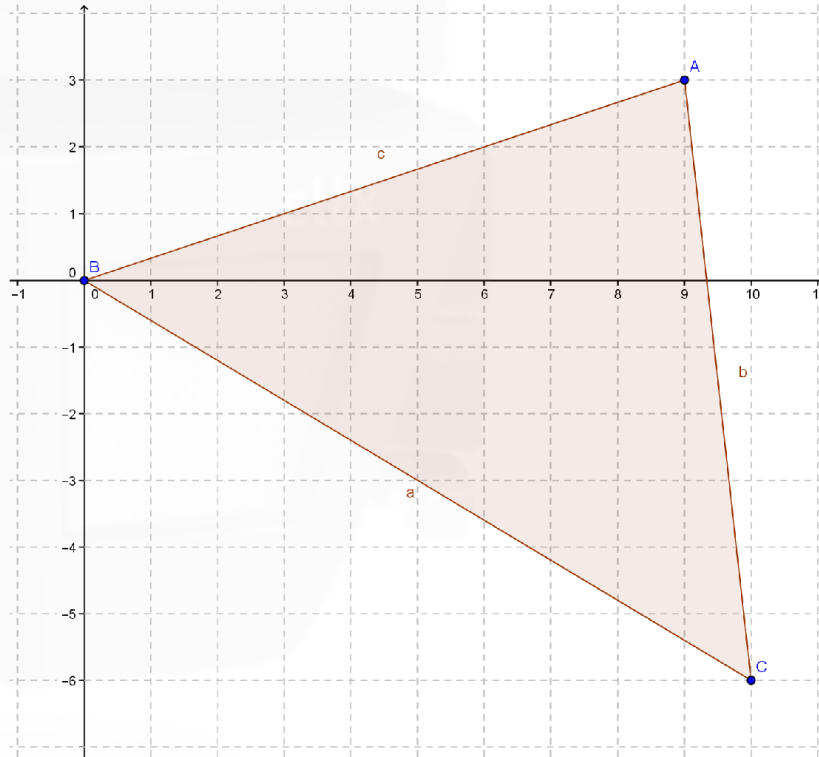
$$\begin{aligned} y - y_1 &= m(x - x_1) & \Leftrightarrow & y - 2 = \frac{1}{3}(x - (-6)) \\ & & \Leftrightarrow & 3y - 6 = x + 6 \\ & & \Leftrightarrow & 3y - x = 12 \end{aligned}$$

(c) Find the area of the triangle ABC .

We wish to translate each of the points so that one of them is on the origin. We will move B to the origin. To do this, we subtract -6 from each x -coordinate and subtract 2 from each y -coordinate. This will leave the area of the triangle unchanged. Thus, we have

$$\begin{aligned} A' &= A - B = (3, 5) - (-6, 2) = (9, 3) \\ B' &= B - B = (-6, 2) - (-6, 2) = (0, 0) \\ C' &= C - B = (4, -4) - (-6, 2) = (10, -6) \end{aligned}$$

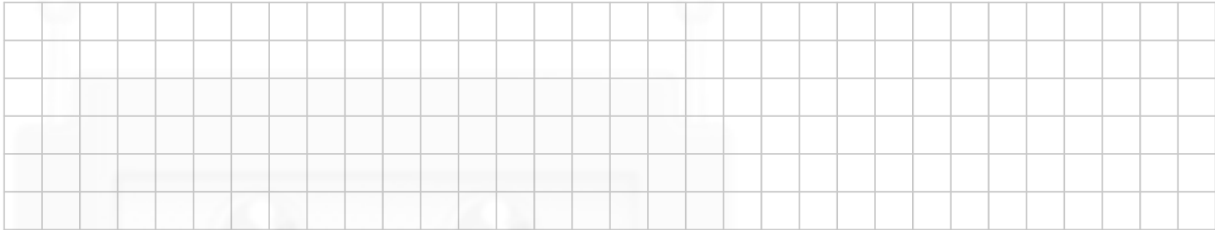
Now, our points look like this:



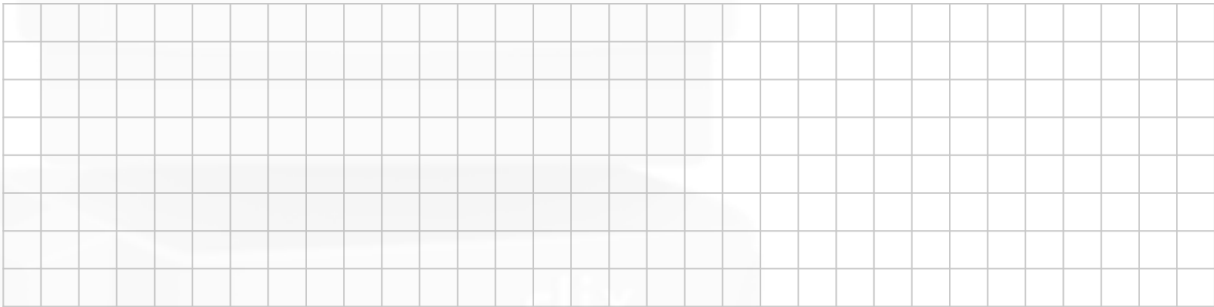
The formula for the area of a triangle with one corner at the origin and the other two at (x_1, y_1) and (x_2, y_2) is $A = \frac{1}{2} |x_1 y_2 - x_2 y_1|$, so the area of $A'B'C'$, i.e. the area of ABC is

$$A = \frac{1}{2} |(9)(-6) - (10)(3)| = \frac{1}{2} |-54 - 30| = \frac{1}{2} |-84| = 42 \text{ units}^2$$

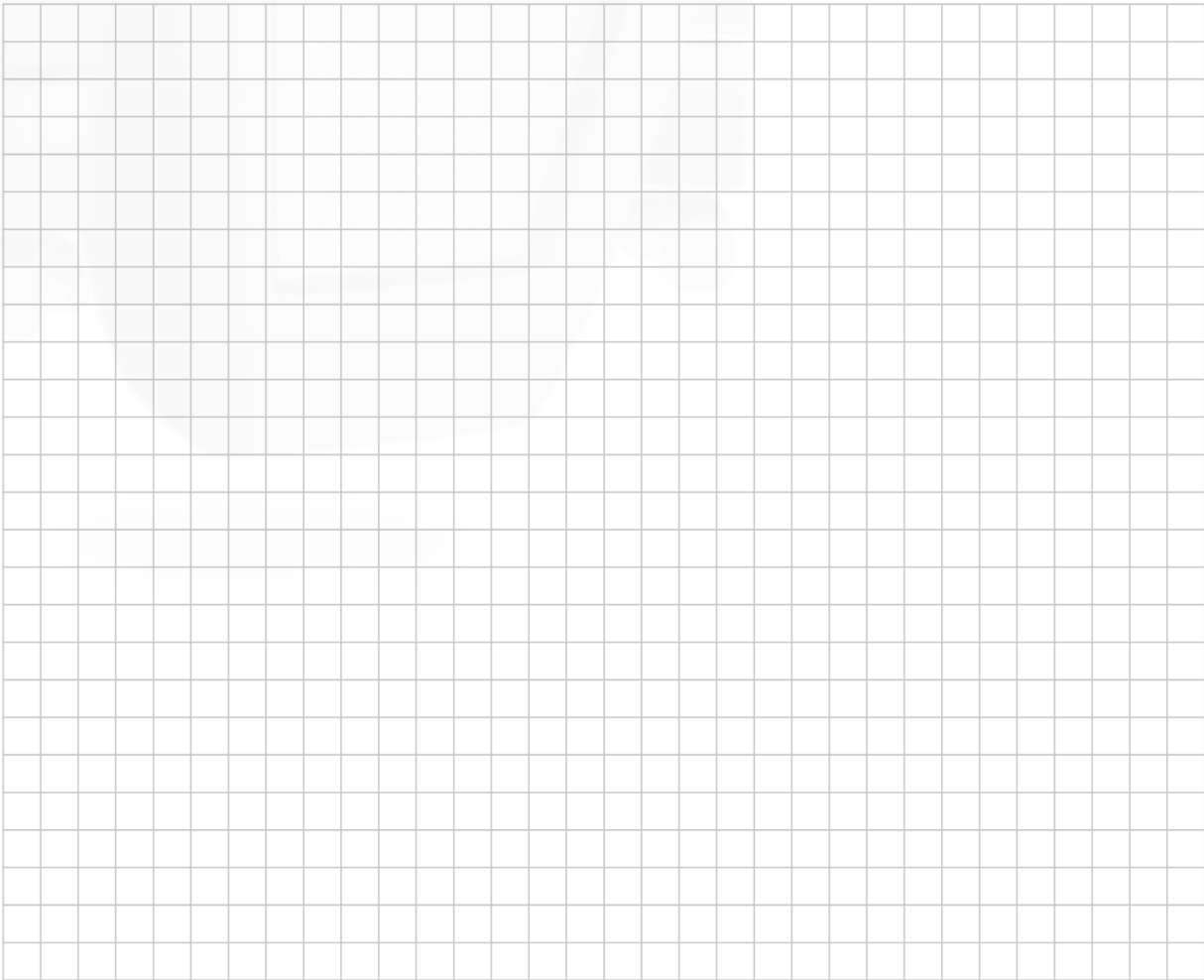
(a) l is the line $3x + 2y + 18 = 0$. Find the slope of l .



(b) The line k is perpendicular to l and cuts the x -axis at the point $(7, 0)$. Find the equation of k .



(c) Find the co-ordinates of the point of intersection of the lines l and k .



- (a) l is the line $3x + 2y + 18 = 0$. Find the slope of l .

$$3x + 2y + 18 = 0 \Rightarrow 2y = -3x - 18 \Rightarrow y = -\frac{3}{2}x - 9$$

$$\text{Slope} = -\frac{3}{2}$$

- (b) The line k is perpendicular to l and cuts the x -axis at the point $(7, 0)$. Find the equation of k .

$$k \perp l \Rightarrow m \times -\frac{3}{2} = -1 \Rightarrow m = \frac{2}{3}$$

$$y - y_1 = m(x - x_1)$$

$$k: y - 0 = \frac{2}{3}(x - 7) \Rightarrow 3y = 2x - 14 \Rightarrow 2x - 3y - 14 = 0$$

- (c) Find the co-ordinates of the point of intersection of the lines l and k .

$$\begin{array}{rcl} 3x + 2y = -18 & \Rightarrow & 9x + 6y = -54 \\ 2x - 3y = 14 & & \underline{4x - 6y = 28} \\ & & 13x = -26 \Rightarrow x = -2 \end{array}$$

$$3x + 2y = -18 \Rightarrow 3(-2) + 2y = -18 \Rightarrow 2y = -12 \Rightarrow y = -6$$

$$\text{Co-ordinates: } (-2, -6)$$

Question 3

(25 marks)

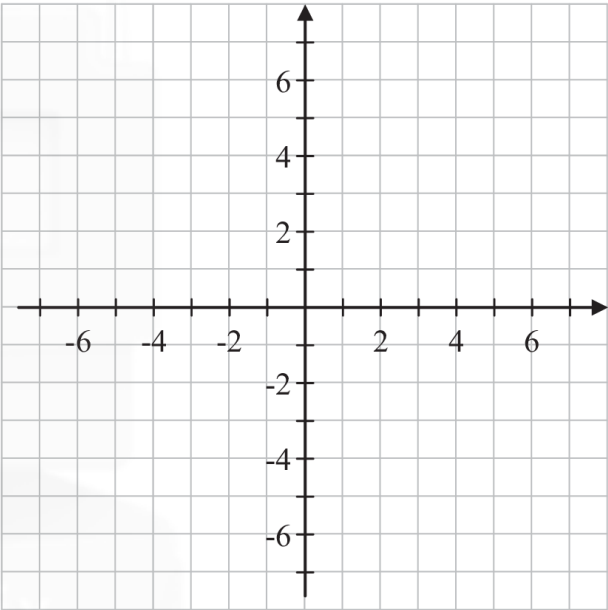
The points A , B , and C have co-ordinates as follows:

$A (3, 5)$

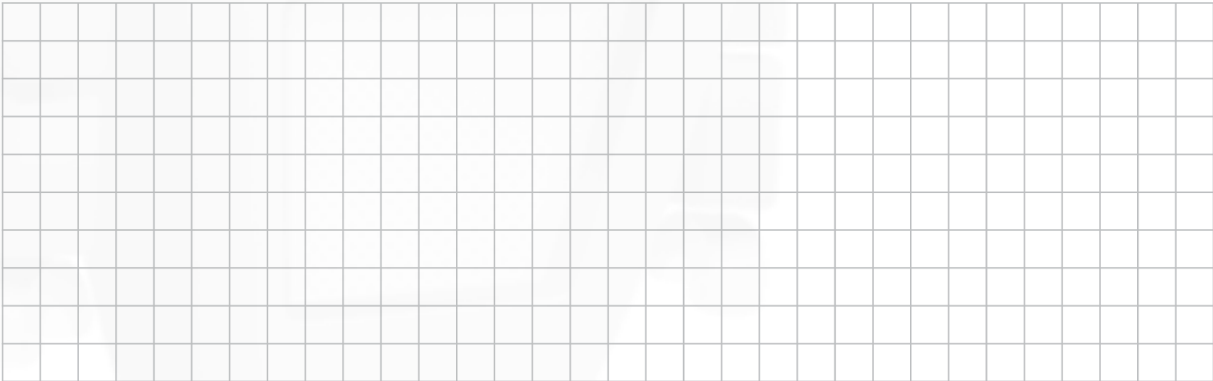
$B (-6, 2)$

$C (4, -4)$

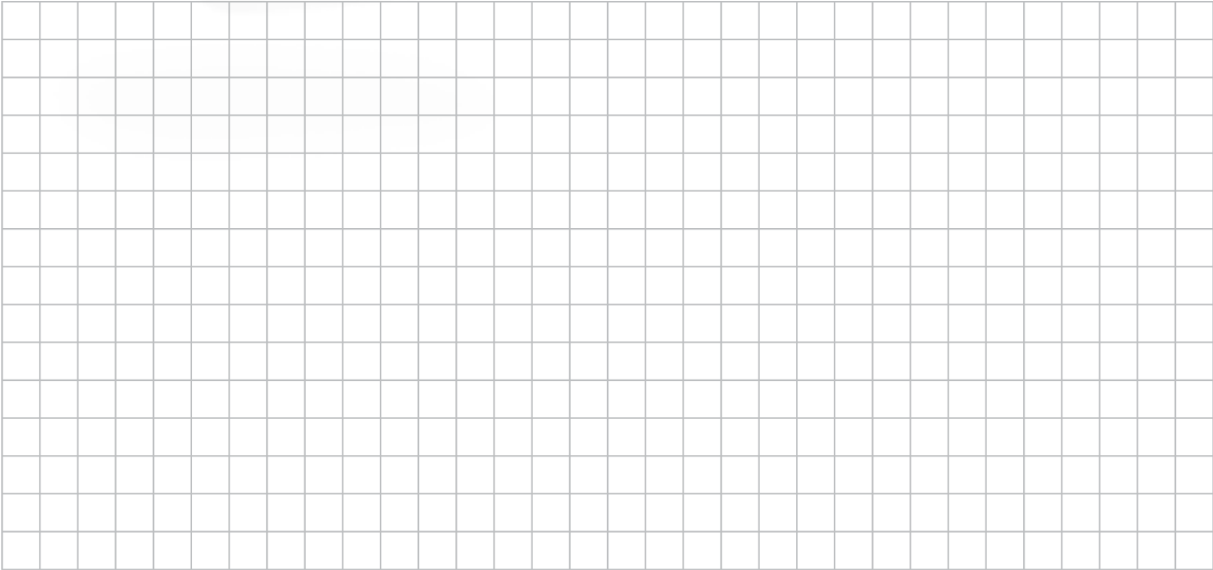
- (a) Plot A , B , and C on the diagram.



- (b) Find the equation of the line AB .



- (c) Find the area of the triangle ABC .



Question 4

(25 marks)

The circle c has centre $P(-2, -1)$ and passes through the point $Q(3, 1)$.

- (a) Show c , P , and Q on a co-ordinate diagram.



- (b) Find the radius of c and hence write down its equation.



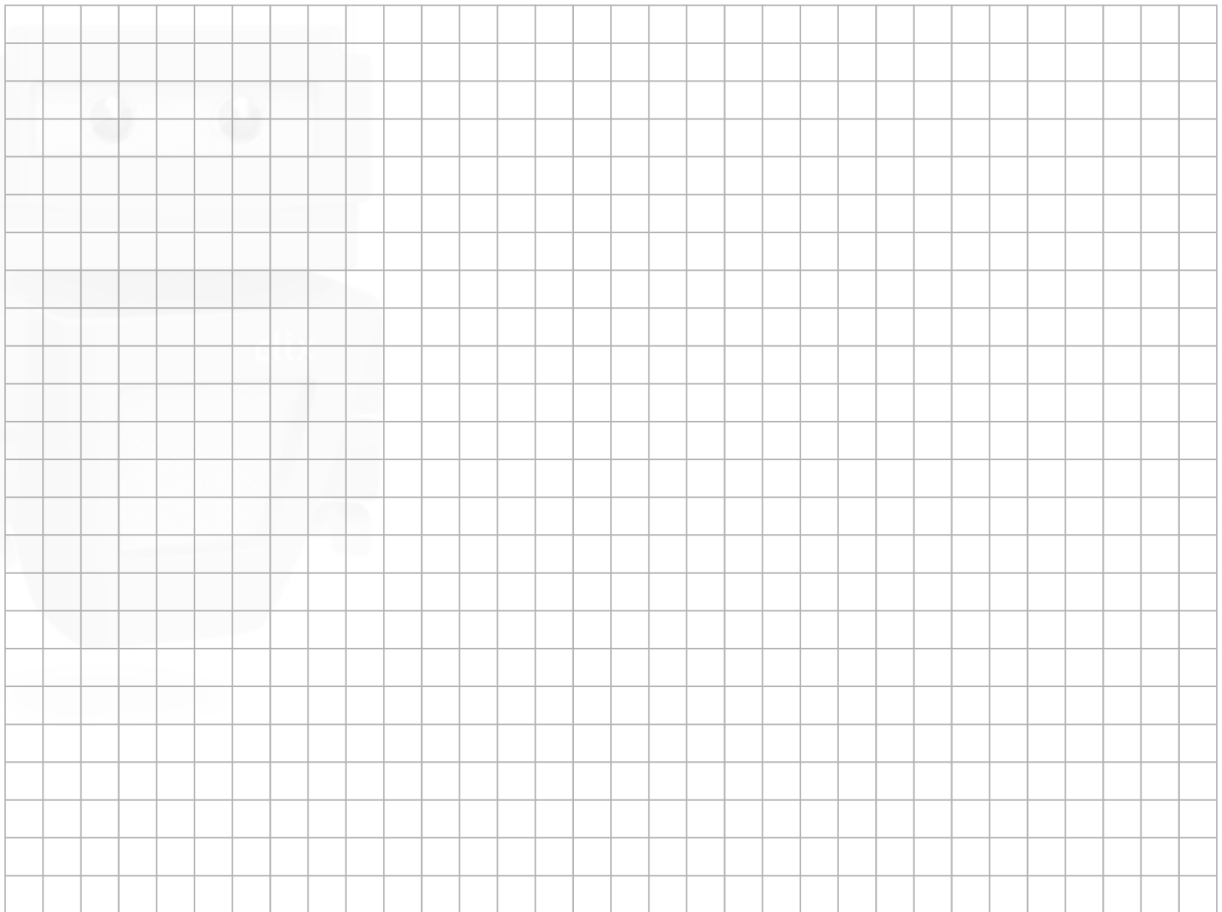
- (c) R is the point $(1, 6)$. By finding the slopes of PQ and QR , show that QR is a tangent to c .

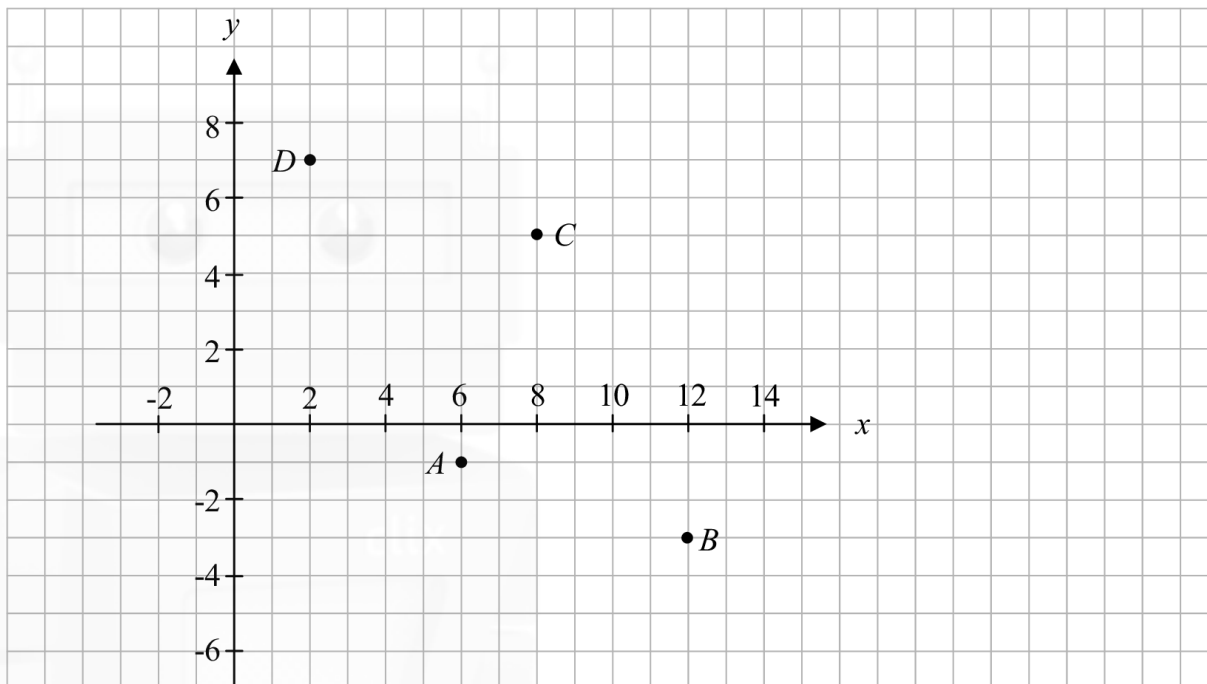


-
- A blank Cartesian coordinate system is shown. The horizontal x-axis is labeled 'x' at its right end and has tick marks at intervals of 2, labeled -2, 2, 4, 6, 8, 10, 12, and 14. The vertical y-axis is labeled 'y' at its top end and has tick marks at intervals of 2, labeled -6, -4, -2, 2, 4, 6, and 8. A grid of light gray lines is present, with major lines every 2 units and minor lines every 1 unit. The origin (0,0) is the intersection of the two axes.

- Second method:

- (c) Use one of the ways you have described to show that $ABCD$ is a parallelogram.





- (b) Describe two different ways of showing, using co-ordinate geometry techniques, that the points form a parallelogram $ABCD$.

Any TWO of:

- Show that opposite sides are parallel by showing the slopes of opposite lines are equal.
- Show that the diagonals bisect each other by showing the midpoint of $[AC]$ equals the midpoint of $[DB]$.
- Show that the opposite sides are equal in length using the length formula.
- Show that $\overrightarrow{AB}$ maps D onto C or similar.

- (c) Use one of the ways you have described to show that $ABCD$ is a parallelogram.

$$\text{Slope } AB = \frac{-3+1}{12-6} = -\frac{2}{6}, \text{ Slope } DC = \frac{7-5}{2-8} = -\frac{2}{6} \Rightarrow AB \parallel DC$$

$$\text{Slope } BC = \frac{5+3}{8-12} = -2, \text{ Slope } AD = \frac{7+1}{2-6} = -2 \Rightarrow BC \parallel AD$$

Hence, $ABCD$ a parallelogram

or

$$\text{Midpoint } [AC] = \left(\frac{6+8}{2}, \frac{-1+5}{2} \right) = (7, 2), \text{ Midpoint } [BD] = \left(\frac{12+2}{2}, \frac{-3+7}{2} \right) = (7, 2),$$

$\Rightarrow$ Diagonals bisect. Hence, $ABCD$ a parallelogram

or

$$\text{Length } [AB] = \sqrt{(12-6)^2 + (-3+1)^2} = \sqrt{40}$$

$$\text{Length } [DC] = \sqrt{(2-8)^2 + (7-5)^2} = \sqrt{40}$$

$$\text{Length } [AD] = \sqrt{(6-2)^2 + (-1-7)^2} = \sqrt{80}$$

$$\text{Length } [BC] = \sqrt{(12-8)^2 + (-3-5)^2} = \sqrt{80}$$

$\Rightarrow$ Opposite sides are equal. Hence, $ABCD$ a parallelogram

or

$$A(6, -1) \rightarrow B(12, -3) \text{ maps } D(2, 7) \rightarrow (2+6, 7-2) = C(8, 5) \text{ or similar}$$

$\Rightarrow \overrightarrow{AB} = \overrightarrow{DC}$. Hence, $ABCD$ a parallelogram

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Question 4

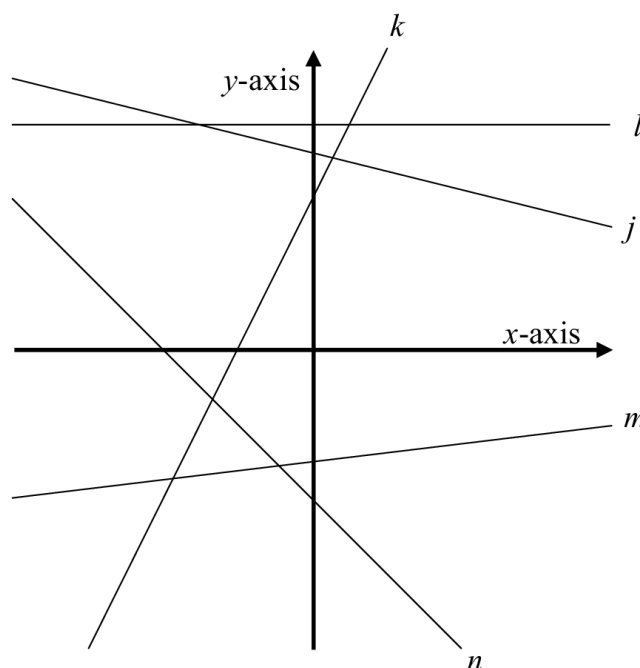
(25 marks)

- (a) Five lines $j, k, l, m,$ and n in the co-ordinate plane are shown in the diagram.

The slopes of the five lines are in the table below.

Complete the table, matching the lines to their slopes.

slope	line
2	
$\frac{1}{8}$	
0	
$-\frac{1}{4}$	
-1	



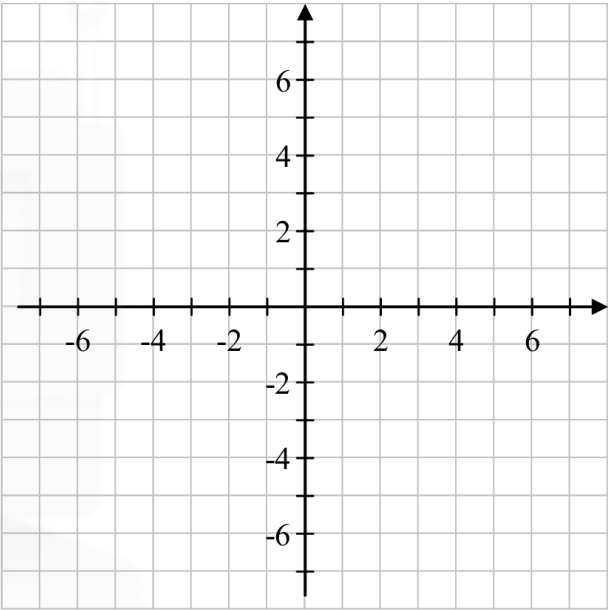
The points A , B , and C have co-ordinates as follows:

$A (3, 5)$

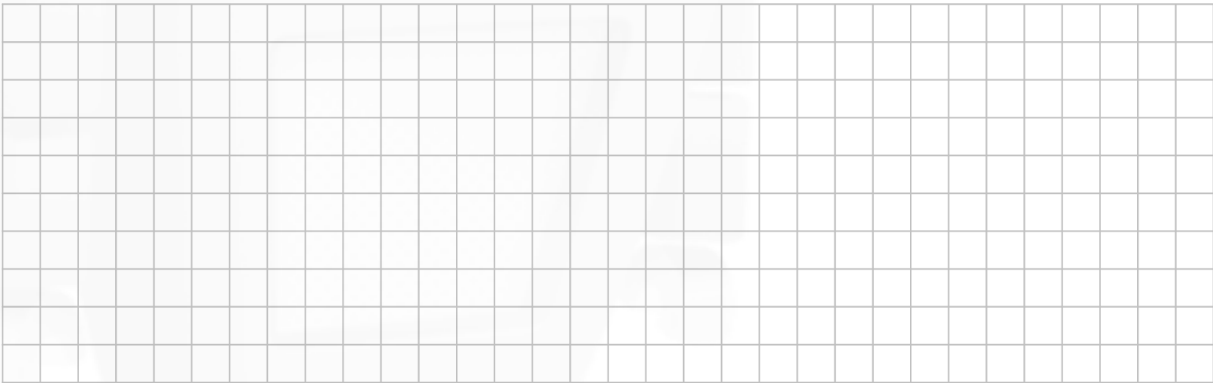
$B (-6, 2)$

$C (4, -4)$

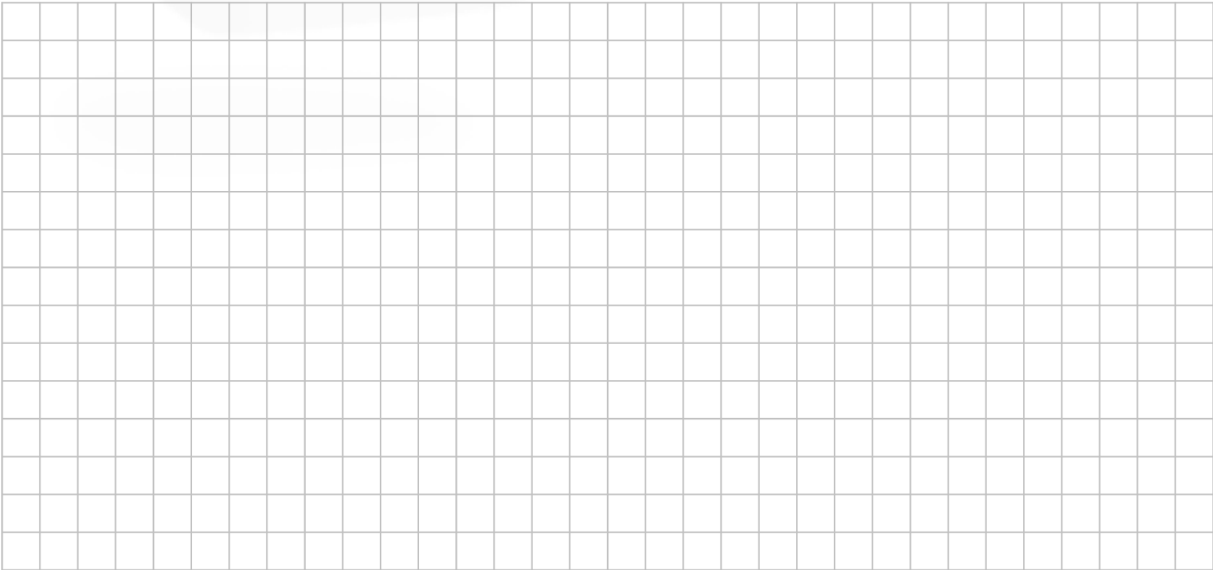
(a) Plot A , B , and C on the diagram.



(b) Find the equation of the line AB .

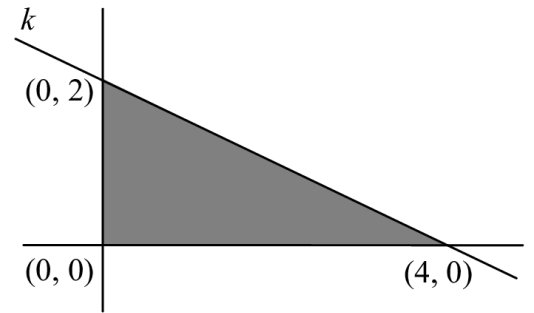


(c) Find the area of the triangle ABC .



- 11. (a)** The line k passes through the points $(0, 2)$ and $(4, 0)$.

- (i)** Find the equation of k .
- (ii)** Write down the three inequalities which define the shaded region in the diagram.



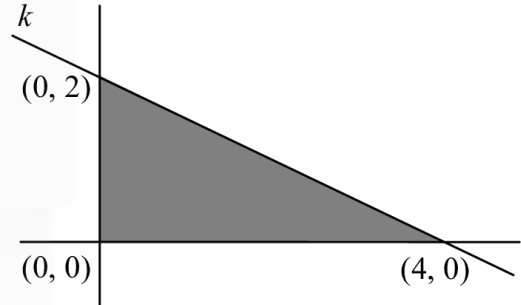
QUESTION 11

Part (a)	15 (10, 5) marks	Att (3, 2)
Part (b)	35 (20, 5, 10) marks	Att (7, 2, 3)

Part (a)	15 (10, 5) marks	Att (3, 2)
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The line k passes through the points $(0, 2)$ and $(4, 0)$.

- (i) Find the equation of k .
- (ii) Write down the three inequalities which define the shaded region in the diagram.



(a) (i)	10 marks	Att 3
----------------	-----------------	--------------

Slope of k $\frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 0}{0 - 4} = \frac{2}{-4} = -\frac{1}{2}$ [4 marks]

Equation of k : $y - y_1 = m(x - x_1)$
 $y - 0 = -\frac{1}{2}(x - 4)$ or $y - 2 = -\frac{1}{2}(x - 0)$ or $[x + 2y - 4 = 0]$. [10 marks]

or

Equation of k : $y = mx + c$
 $0 = -\frac{1}{2}(4) + c \Rightarrow c = 2$. [10 marks]

- * Accept $y - 0 = -\frac{1}{2}(x - 4)$ without work.
- * Apply scheme for Q2, Q3 where relevant.
- * Error in simplifying k to be penalised in section a(ii) subject to marks secured.
- * Treat as TWO blunders when m is not addressed, i.e. $y - 2 = m(x - 0)$ or $y - 2 = (x - 0)$.

Blunders (-3)

- B1 Incorrect relevant formula and continues.
- B2 Mixes up x 's and y 's when substituting.
- B3 $y - 2 = m(x - 0)$ where m is not equal to $-\frac{1}{2}$ without work.
- B4 $y - y_1 = -\frac{1}{2}(x - x_1)$ where (x_1, y_1) is not $(4, 0)$ or $(0, 2)$ without work.
- B5 Point substituted incorrectly.

Slips (-1)

- S1 Numerical slips to a maximum of 3.

Attempts (3 marks)

- A1 Any work of merit.

Worthless (0 marks)

- W1 An arbitrary line without work.

(a) (ii)**5 marks****Att 2**

$$x \geq 0, \quad y \geq 0, \quad x + 2y - 4 \leq 0 \text{ or equivalent.}$$

- * Accept correct inequalities without work.
- * Accept $<$ for $\leq$ and $>$ for $\geq$.
- * Accept an inequality consistent with candidate's k . [See 3rd * a (i)]
- * Award 2 marks for one correct inequality, 4 marks for 2 correct and 5 marks for 3 correct.

Attempts (2 marks)

- A1 Substitutes any point and stops.
- A2 $x \leq 0$ or $y \leq 0$ and stops, (without work).
- A3 Incorrect or no conclusion if working on K e.g. $x + 2y - 4 = 0 \Rightarrow 0 + 2(0) - 4 = 0$.
- A4 Mathematical error in testing a point, e.g. sign error.
- A5 Some relevant step e.g. $x = 0$, k simplified accept even in a(i).
- A6 Lists all cases, i.e. $x \geq 0, x \leq 0, y \geq 0, y \leq 0, x + 2y - 4 \leq 0, x + 2y - 4 \geq 0$

Worthless (0 marks)

- W1 Writes equation of k and stops (without simplifying).
- W2 Draws the given diagram or writes (0, 0), (0, 2), (4, 0) only.

Part (b)**35 (15, 5, 5, 10) marks****Att (5, 2, 2, 3)**

A contractor has the task of loading containers onto a truck. There are two types of containers: heavy containers which weigh 160 kg each and light containers which weight 40 kg each. The truck can carry, at most, a total weight of 2080 kg.

The time taken to load a heavy container is 3 minutes. The time taken to load a light container is 2 minutes. The total time spent loading a truck cannot be greater than 54 minutes.

- (i) Taking x as the number of heavy containers and y as the number of light containers, write down two inequalities in x and y and illustrate these on graph paper.
- (ii) The contractor charges €48 to load each heavy container and €36 to load each light container. How many of each should be loaded in order to maximise income?
- (iii) On your graph, show the region where the income is at most €576.

(b) (i) Inequalities**15 (10, 5) marks****Att (3, 2)**

Weight: $160x + 40y \leq 2080$ or $[4x + y \leq 52]$
 Time: $3x + 2y \leq 54$.

- * Accept correct multiples or fractions of inequalities or the use of different letters.
- * Apply (-3), once, if no inequality sign or the incorrect inequality sign is written the first time it occurs.
- * Accept $<$ for $\leq$.
- * Better inequality marked out of 10. Other inequality marked out of 5.

	Heavy / x	Light / y	Maximum
Weight	160	40	2080
Time	3	2	54

12 Marks*Blunders (-3)*

- B1 Mixes up x 's and y 's (once if consistent error).
- B2 Confuses rows and columns in table, e.g. $160x + 3y \leq 2080$ (once if consistent).
- B3 Decimal blunder applies for error with zeros in equation, unless an obvious misreading.

Attempts (3 or 2 marks)

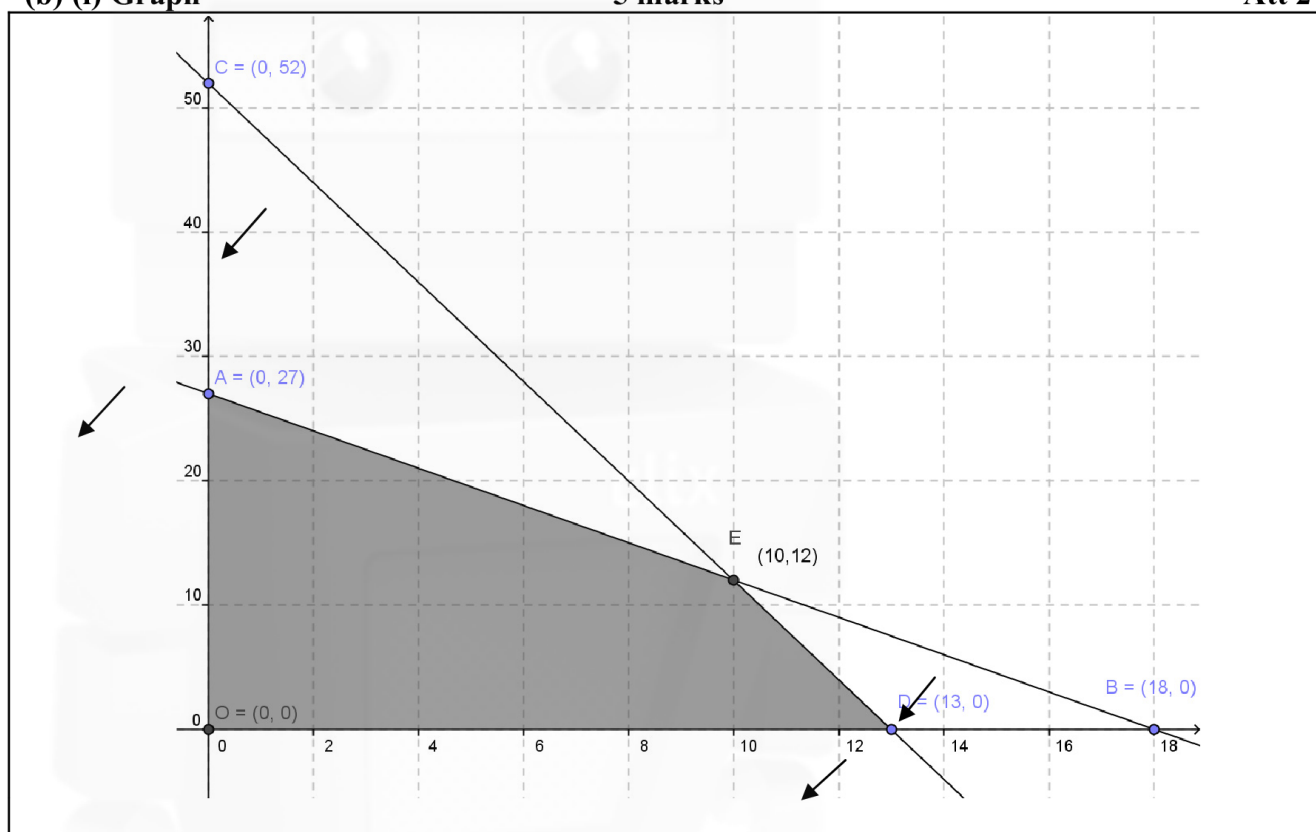
A1 Incomplete relevant data in table and stops, e.g. $160x$ or $3x$ or $40y$ or $2y$ or ≤ 2080 or ≤ 54 (each inequality).

A2 Any other correct inequality, e.g. $x \geq 0$, $y \geq 0$, (each time).

(b) (i) Graph

5 marks

Att 2



- * Points or scales required.
- * Correct shading over-rides arrows or correct arrows overrule shading.
- * Inequalities not written but correct graph drawn – award 0 + 5 marks.
- * Two lines drawn and no shading indicated, only one of the following cases applies:

Case 1: Two sets of arrows in expected direction	5 marks
Case 2: Two sets of arrows in unexpected direction	5 marks
Case 3: One set of arrows “correct”, the other “incorrect”	2 marks
Case 4: One line with and the other without arrows	2 marks
Case 5: No arrows	2 marks
Case 6: Half-planes consistent with incorrect, penalised inequalities.	5 marks

Blunders (-3)

B1 Blunder in plotting a line or calculations.

B2 Incorrect shading, e.g. one or both of the small triangles shaded, (subject to case 6 above).

B3 Vertical: x -axis, Horizontal: y -axis.

Attempts (2 marks)

A1 Some relevant work towards a point on a line.

A2 Draws scaled axes or axes and one line.

(b) (ii) Intersection/Income**5 marks****Att 2**

$$4x + y = 52$$

$$8x + 2y = 104$$

$$\underline{3x + 2y = 54} \Rightarrow 3x + 2y = 54$$

$$\underline{5x = 50} \Rightarrow x = 10 \quad \text{and} \quad y = 12$$

Step 1	Vertices	$48x + 36y$	Income
	(0, 0)	0 + 0	0
Step 2	(13, 0)	624 + 0	624
Step 3	(10, 12)	480 + 432	912
Step 4	(0, 27)	0 + 972	972

Step 5 0 heavy containers and 27 light containers to maximise income.

- * Accept candidate's own equations from previous sections.
- * If solving incorrect equations, the point found may be outside the feasible set – award marks for correct work and accept in later sections.
- * Information does not have to be in table form.
- * Accept any correct multiple or fraction of $48x + 36y$ here (may be implied).
- * Accept work on a feasible set of points formed by axes and one line without further penalty.
- * Accept only vertices consistent with previously accepted work, not arbitrary ones.
- * If (18, 0) or (0, 52) is tested and result is used to give maximum income, award attempt 2.
- * Step 5 must be explicitly written to gain full marks.
- * Testing only (0, 27) to get 972 merits Att 2 even if the candidate writes 0 heavy containers and 27 light containers to maximise income, i.e. no comparison means the attempt mark at most.

Blunders (-3)

- B1 Fails to multiply / divide both sides of equation(s) correctly when eliminating variable.
- B2 x or y value only found.

Attempts (2 marks)

- A1 Correct or consistent point of intersection without work or from a graph.
[Should get the *exact same* values from graph as if they had been found algebraically.]
- A2 Any relevant step towards solving equations.
- A3 Any relevant work involving x or y and/or 48, 36 or similar.
- A4 Any attempt at substituting co-ordinates into some relevant expression.
- A5 Any step omitted, subject to the case for awarding 4 marks.

Worthless (0 marks)

- W1 Incorrect point of intersection without work and inconsistent with the lines, subject to marks secured.
- W2 Writing €48 or €36 without further work, subject to marks secured.

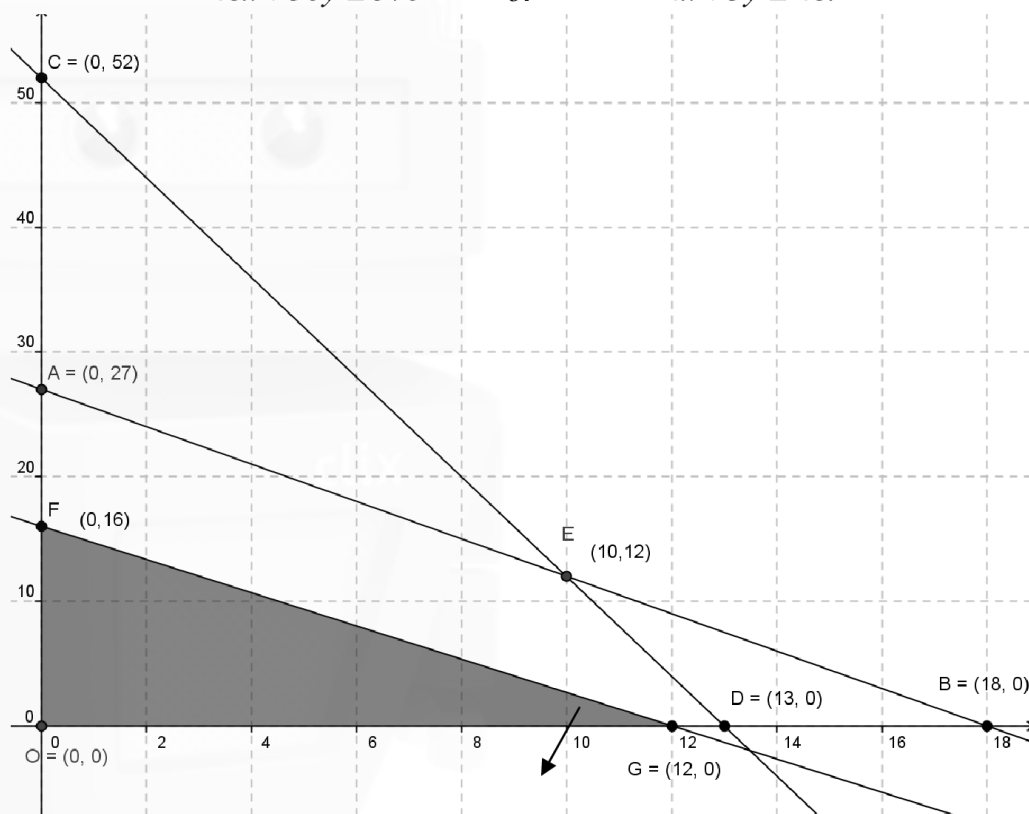
Award marks as follows:

- 5 marks: Answer is fully correct or consistent.
- 4 marks: The maximum value is identified but step 5 not stated.
- 2 marks: Some relevant work. e.g point of intersection.
- 0 marks: Worthless work.

$$48x + 36y \leq 576$$

or

$$4x + 3y \leq 48.$$



- * Accept correct multiples or fractions of the inequality.
- * No penalty for writing $48x + 36y = 576$, provided correct area is indicated.

Award only marks as follows:

10 marks: Answer is fully correct or consistent, with conclusion with expected shading clearly indicated.

3 marks: Some relevant work.

0 marks: Worthless work.

Attempts (3 marks)

A1 The point (12, 0) or (0, 16) plotted.

A2 A subset of the feasible set bounded by the axes shaded.

2. (a) Find the area of the triangle with vertices $(0, 0)$, $(8, -6)$ and $(-1, 5)$.

(b) l is the line $3x - 4y - 15 = 0$.

(i) Verify that $(1, -3)$ is a point on l .

(ii) l intersects the x -axis at P . Find the co-ordinates of P .

The line k passes through the point $(1, -3)$ and is perpendicular to l .

(iii) Show the lines l and k on a co-ordinate diagram.

(iv) Find the equation of k .

(c) $A(2, -1)$ and $B(-4, 7)$ are two points.

(i) Find $|AB|$.

(ii) Find C , the image of B under the translation $(2, -1) \rightarrow (-7, 11)$.

(iii) Show that $|AB| : |AC| = 2 : 5$.

QUESTION 2

Part (a)	5 marks	Att 2
Part (b)	25 (10, 5, 5, 5) marks	Att (3, 2, 2, 2)
Part (c)	20 (10, 5, 5) marks	Att (3, -, -)

Apply the following to each section of question 2 and question 3.

If the correct formula is not written, any sign or substitution error is at least a blunder.

Blunders (-3)

- B_a Two or more incorrect substitutions.
- B_b Switches x and y in substituting or treats as a pair of couples (x_1, x_2) and (y_1, y_2) .
- B_c Error in the central sign in a formula.

Slips (-1)

- S_a One incorrect non-central sign.
- S_b One incorrect substitution in the formula.
- S_c Obvious misreading of one co-ordinate.

Attempts

- A_a The correct relevant formula written and stops. [If not transcribed from tables.]
- A_b The co-ordinates of a relevant point written with x_1 and y_1 identified.
- A_c A correct substitution into relevant formula and stops.

Worthless (0 Marks)

- W_a Correct formula transcribed from tables and stops.

Part (a)	5 marks	Att 2
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Find the area of the triangle with vertices $(0, 0)$, $(8, -6)$ and $(-1, 5)$.

(a)	5 marks	Att 2
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$(0, 0)$, $(8, -6)$ and $(-1, 5)$

$$\text{Area} = \frac{1}{2} |x_1 y_2 - x_2 y_1| = \frac{1}{2} |8 \times 5 - (-1)(-6)| = \frac{1}{2} |40 - 6| = \frac{1}{2} |34| = 17.$$

or

$$\begin{aligned} \text{Area} &= \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \downarrow_{2 \text{ marks}} \\ &= \frac{1}{2} |0(-6 - 5) + 8(5 - 0) + (-1)(0 + 6)| = \frac{1}{2} |0 + 40 - 6| = \frac{1}{2} |34| = 17 \end{aligned}$$

or

$$\begin{aligned} \text{Area} &= \frac{1}{2} [x_1 y_2 + x_2 y_3 + x_3 y_1 - x_1 y_3 - x_3 y_2 - x_2 y_1] \downarrow_{2 \text{ marks}} \\ &= \frac{1}{2} |0 \times -6 + 8 \times 5 + (-1) \times 0 - 0 \times 5 - (-1)(-6) - 8 \times 0| \\ &= \frac{1}{2} |0 + 40 + 0 - 6 + 0| = 17. \end{aligned}$$

* $\frac{1}{2} |-34| = -17$ incurs no penalty.

Blunders (-3)

- B1 Incorrect relevant formula and continues e.g. $\frac{1}{2} |x_1 y_2 + x_2 y_1|$ or omits the $\frac{1}{2}$.
- B2 An incorrect translation or required translation omitted.

Attempts (2 marks)

- A1 Uses the distance formula or the perpendicular distance formula, (at least one substitution).
- A2 Plots one or more points, to the eye.
- A3 A correct answer without work.

Worthless (0 marks)

- W1 Irrelevant formula and stops e.g. $\frac{1}{2}$ on its own.

Part (b)**25 (10, 5, 5, 5) marks****Att (3, 2, 2, 2)**

l is the line $3x - 4y - 15 = 0$.

(i) Verify that $(1, -3)$ is a point on l .

(ii) l intersects the x -axis at P . Find the co-ordinates of P .

The line k passes through the point $(1, -3)$ and is perpendicular to l .

(iii) Show the lines l and k on a co-ordinate diagram.

(iv) Find the equation of k .

(b) (i)**10 marks****Att 3**

$$3x - 4y - 15 = 0 \Rightarrow 3(1) - 4(-3) - 15 = 3 + 12 - 15 = 15 - 15 = 0$$

[Hence, $(1, -3) \in l$.]

* Accept consistent answers in this and subsequent sections.

Blunders (-3)

B1 Substitution, but work not completed to arrive at LHS = RHS.

Slips(-1)

S1 No conclusion if L.H.S. $\neq$ R.H.S.

Attempts (3 marks)

A1 Some substitution attempted or some correct work with the equation. e.g. $3x - 4y = 15$.

A2 Plots $(1, -3)$ for this section.

(b) (ii)**5 marks****Att 2**

$$3x - 4y - 15 = 0$$

$$y = 0 \Rightarrow 3x - 4(0) - 15 = 0 \Rightarrow 3x = 15 \Rightarrow x = 5.$$

[$P(5, 0)$]

* Accept a correct answer without work.

Blunders (-3)

B1 Finds intercept on the y -axis. i.e. $(0, -3\frac{3}{4})$.

Attempts (2 marks)

A1 Some relevant step e.g. writes $y = 0$ and stops.

A2 Finds a random point on the line.

A3 Writes $(*, 0)$.

A4 Attempts to write in form $y = mx + c$.

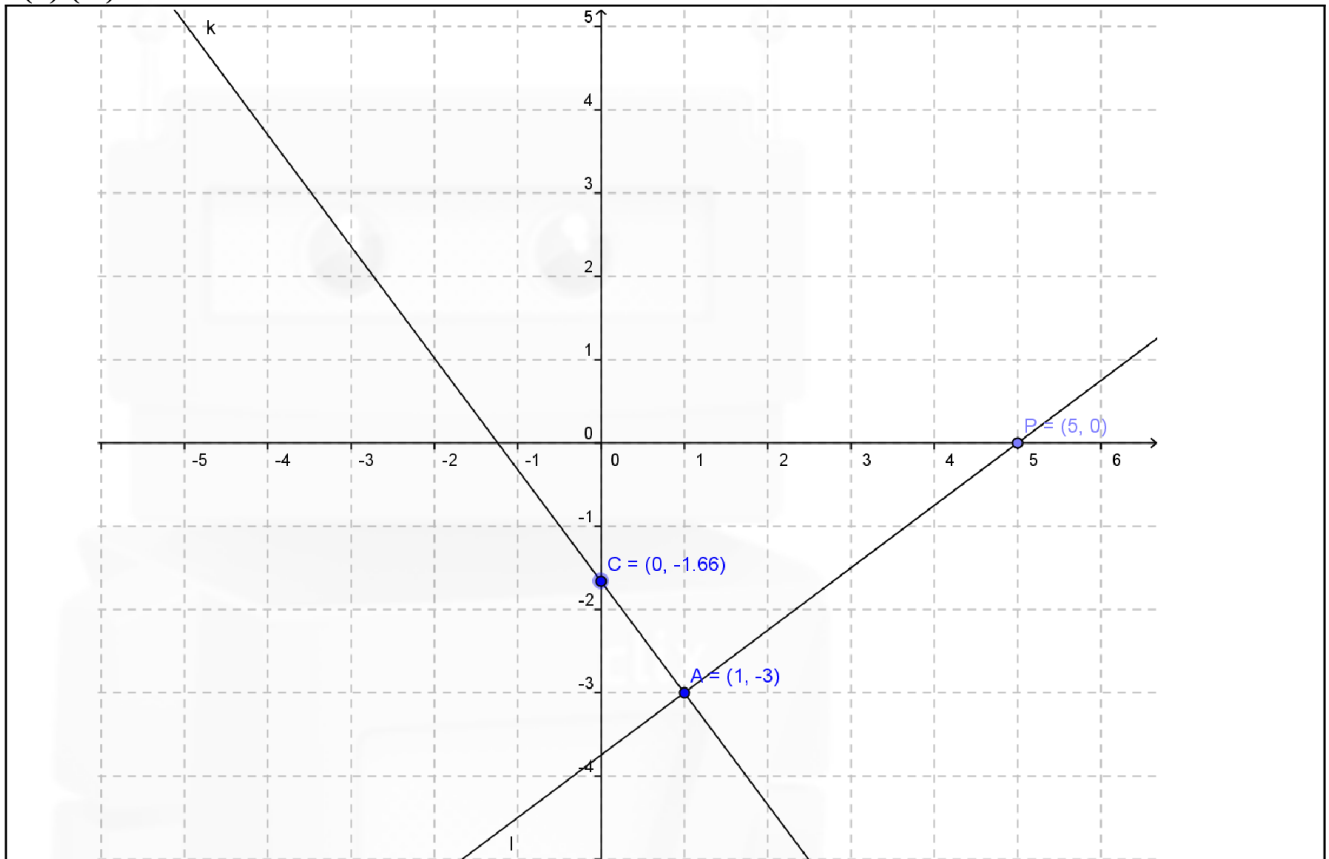
Worthless (0 marks)

W1 Writes $x = 0$ and stops.

(b) (iii)

5 marks

Att 2



- * Accept candidate's co-ordinates of P from (ii) and/or candidate's equation of k from (iv).
- * Intervals should be indicated or implied.
- * If (ii) is not answered but $P(5, 0)$ is labelled on the graph award 5 marks for (ii).
- * Work must be shown if diagram is not consistent with (ii).
- * Accept tolerance of $\pm 10^\circ$ in perpendicular.

Blunders (-3)

- B1 Scales unreasonably inconsistent, (to the eye).
- B2 Different scales on x and y axes, (to the eye).
- B3 Measure of angle between l and k outside tolerance of $\pm 10^\circ$.
- B4 k does not pass through $(1, -3)$.
- B5 Plots $P(5, 0)$ on the y -axis.
- B6 Points plotted but not joined.
- B7 Vertical x -axis.

Attempts (2 marks)

- A1 Draws scaled axes and stops, [for this section b(iii)].
- A2 Plots one correct point or just l and stops.

Worthless (0 marks)

- W1 Draws arbitrary line l with k not perpendicular to l , subject to A1.

(b) (iv)

5 marks

Att 2

$$l: 3x - 4y - 15 = 0 \Rightarrow 4y = 3x - 15 \Rightarrow y = \frac{3}{4}x - \frac{15}{4}.$$

Hence slope of $l = \frac{3}{4}$.

$$\frac{3}{4} \times m = -1 \Rightarrow m = -\frac{4}{3}, \text{ the slope of } k.$$

$$y - y_1 = m(x - x_1) \Rightarrow y + 3 = -\frac{4}{3}(x - 1) \downarrow_{5 \text{ marks}} \Rightarrow [3y + 9 = -4x + 4 \Rightarrow 4x + 3y + 5 = 0].$$

or

$$l: 3x - 4y - 15 = 0.$$

$$k \perp l \Rightarrow k: 4x + 3y + c = 0$$

$$(1, -3) \in k \Rightarrow 4(1) + 3(-3) + c = 0 \Rightarrow c = 5 \downarrow_{5 \text{ marks}}$$

$$[k: 4x + 3y + 5 = 0]$$

or

$$y = mx + c$$

$$k: y = -\frac{4}{3}x + d$$

$$(1, -3) \in k \Rightarrow -3 = -\frac{4}{3} + d = 0 \Rightarrow -9 = -4 + 3d \Rightarrow d = -\frac{5}{3} \downarrow_{5 \text{ marks}}$$

$$[y = -\frac{4}{3}x - \frac{5}{3} \Rightarrow 3y = -4x - 5 \text{ or } 4x + 3y + 5 = 0].$$

* Allow equation of k work to precede graphing.

Blunders (-3)

B1 Incorrect slope of k , i.e. $m_1 m_2 \neq -1$, e.g. $\frac{4}{3}$ or $-\frac{3}{4}$.

B2 Use of $m_1 m_2 = -1$ omitted or applied incorrectly.

B3 Uses an arbitrary point.

Misreadings (-1)

M1 k passes through $P(5, 0)$.

Attempts (2 marks)

A1 Correct relevant formula not transcribed from tables and stops e.g. $m_1 m_2 = -1$ or $m = -\frac{a}{b}$.

A2 k written as $4x + 3y$ and stops.

A3 Transposes x or y and stops.

A4 Correct answer without work.

Part (c)**20 (10, 5, 5) marks****Att (3, -, -)** $A(2, -1)$ and $B(-4, 7)$ are two points.**(i)** Find $|AB|$.**(ii)** Find C , the image of B under the translation $(2, -1) \rightarrow (-7, 11)$.**(iii)** Show that $|AB| : |AC| = 2 : 5$.**(c) (i)****10 marks****Att 3** $A(2, -1)$ and $B(-4, 7)$

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-4 - 2)^2 + (7 + 1)^2} \downarrow_{4\text{marks}} = \sqrt{36 + 64} = \sqrt{100} \downarrow_{7\text{marks}} = 10$$

(c) (ii)**5 marks****Hit or miss**Translation $(2, -1) \rightarrow (-7, 11) \Rightarrow (-9, 12)$ maps $B(-4, 7) \rightarrow C(-4 - 9, 7 + 12) = C(-13, 19)$.

* Accept correct answer without work.

Award only these marks in section (ii)

5 marks Correct answer.

Otherwise 0 marks.

(c) (iii)**5 marks****Hit or miss** $A(2, -1)$ and $C(-13, 19)$

$$|AC| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-13 - 2)^2 + (19 + 1)^2} = \sqrt{225 + 400} = \sqrt{625} = 25$$

$$|AB| : |AC| = 10 : 25 = [2 : 5].$$

Award only these marks in section (iii)

5 marks Correct answer.

Otherwise 0 marks

Blunders (-3)B1 Incorrect formula, e.g. $\sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$ or $\sqrt{(x_2 + x_1)^2 + (y_2 + y_1)^2}$ and continues.B2 Mathematical error, e.g. Incorrect use of $\sqrt{}$ B3 Stops at $\sqrt{100}$ in part c(i).**Attempts (3 marks)**A1 Oversimplifies formula, e.g. $\sqrt{(x_2 - x_1) + (y_2 - y_1)}$ with some correct substitution.

A2 Correct answer without work in c (i).

3. (a) A circle with centre $(0, 0)$ passes through the point $(5, -12)$.
- (i) Find the radius of the circle.
 - (ii) Write down the equation of the circle.
- (b) The circle c has equation $x^2 + y^2 = 17$.
 l is the line $x - 4y - 17 = 0$.
The line l is a tangent to c at the point T .
- (i) Find the co-ordinates of T .
 - (ii) The point T is one end-point of a diameter of c .
Find the co-ordinates of the other end-point.
- (c) A circle has equation $x^2 + (y - 7)^2 = 100$.
- (i) Write down the co-ordinates of the centre of the circle and the radius of the circle.
 - (ii) The point $(6, h)$ is on the circle. Find the two possible values of h .

QUESTION 3

Part (a)	10 (5, 5) marks	Att (2, 2)
Part (b)	20 (15, 5) marks	Att (5, 2)
Part (c)	20 (10, 10) marks	Att (3, 3)

Part (a)	10 (5, 5) marks	Att (2, 2)
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A circle with centre (0, 0) passes through the point (5, -12).

- (i) Find the radius of the circle.
(ii) Write down the equation of the circle.

(a) (i)	5 marks	Att 2
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$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(5 - 0)^2 + (-12 - 0)^2} = \sqrt{25 + 144} = \sqrt{169} \downarrow_{2 \text{ marks}} = 13.$$

or

$$x^2 + y^2 = r^2 \downarrow_{3 \text{ marks}} \Rightarrow (5)^2 + (-12)^2 = r^2 \Rightarrow 25 + 144 = r^2 \Rightarrow r^2 = 169 \downarrow_{2 \text{ marks}} \Rightarrow r = 13$$

* Accept correct answer without work.

(a) (ii)	5 marks	Att 2
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$$x^2 + y^2 = 13^2 = 169.$$

* Accept answer based on candidates work in a (i).

* Accept correct answer without work.

Blunders (-3)

- B1 Incorrect relevant formula with substitution, e.g. $x^2 - y^2 = r^2$ (ii).
B2 Leaves radius as $\sqrt{169}$ (i).
B3 Mathematical error.
B4 No squares or misuse of Pythagoras.
B5 $x^2 + y^2 = 13$ or 26 (ii).
B6 Incorrect or inconsistent centre or radius, e.g. centre = (5, -12).

Attempts (2 marks)

- A1 A formula with $(x_2 - x_1)$ or $(y_2 - y_1)$ and some correct substitution.
A2 Statement of reference to or some use of Pythagoras, e.g. 5^2 .
A3 Any work of merit, e.g. plots (5, -12) correctly, (x or y-axis scaled).
A4 Writes $x^2 + y^2 = r^2$.
A5 Statement that radius is the distance from centre to point (5, -12).
A6 Plots and identifies (0, 0) and (5, -12). (i).

Worthless (0 marks)

- W1 $5 + 12 = 17$ (i)

The circle c has equation $x^2 + y^2 = 17$.

l is the line $x - 4y - 17 = 0$.

The line l is a tangent to c at the point T .

- (i) Find the co-ordinates of T .
 (ii) The point T is one end-point of a diameter of c .
 Find the co-ordinates of the other end-point.

(b) (i)

15 marks

Att 5

I

$$\begin{aligned} x - 4y - 17 = 0 &\Rightarrow x = 4y + 17 \rightarrow x^2 + y^2 = 17 \Rightarrow (4y + 17)^2 + y^2 = 17 \\ &\Rightarrow 16y^2 + 136y + 289 + y^2 - 17 = 0 \Rightarrow 17y^2 + 136y + 272 \Rightarrow y^2 + 8y + 16 \\ &\Rightarrow (y + 4)^2 = 0 \Rightarrow y = -4 \rightarrow x = 4y + 17 \Rightarrow x = 4(-4) + 17 = -16 + 17 = 1 \quad [\text{Pt}(1, -4)] \end{aligned}$$

or

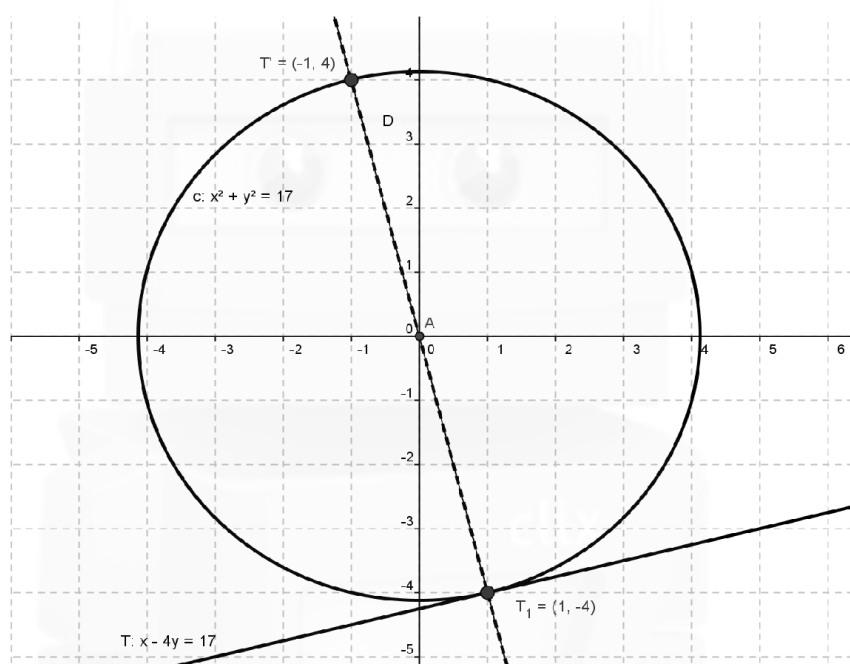
$$y = \frac{-136 \pm \sqrt{18496 - 18496}}{34} = \frac{-136}{34} = -4 \Rightarrow x = 4(-4) + 17 = -16 + 17 = 1 \quad [\text{Pt}(1, -4)]$$

II

$$xx_1 + yy_1 = r^2 \Rightarrow (1)x - 4y = 17 \Rightarrow x_1 = 1 : y_1 = -4$$

* Accept the correct point verified correctly in **both** line and circle equations.

Case III



$$m_l = \frac{1}{4}$$

$$\Rightarrow m_{\text{radius}} = -4$$

Equation of radius

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 0 = -4(x - 0)$$

$$\Rightarrow 4x + y = 0$$

Solve

$$4x + y = 0 \quad \times 4$$

$$x - 4y = 17$$

$$16x + 4y = 0$$

$$x - 4y = 17$$

$$17x = 17$$

$$\Rightarrow x = 1$$

$$\Rightarrow 4(1) + y = 0$$

$$\Rightarrow y = -4$$

$$[T = (1, -4)]$$

Note:

Correct point without work merits **5 marks**.

Award only these marks in section (b)(i):

15 marks: Correct answer with work.

5 marks: Some work of merit, otherwise 0 marks.

Attempts (5)

A1 States centre $(0, 0)$ or $r = \sqrt{17}$ or $r^2 = 17$.

A2 Correct answer without work.

A3 Correct graphical solution. [Note *]

A4 Any correct relevant work. e.g. $x - 4y = 17$.

A5 Plots l and/or c and stops.

A6 Finds only one co-ordinate with work.

(b) (ii)**5 marks****Att 2**

$$(1, -4) \rightarrow (0, 0) \rightarrow (-1, 4)$$

or

$$\frac{1+x}{2}=0; \frac{-4+y}{2}=0 \Rightarrow x=-1, y=4 \rightarrow (-1, 4)$$

* Accept a correct answer without work shown.

* Accept answer consistent with work in b(i).

Blunders (-3)

B1 Error in use of translation or central symmetry.

B2 Find only one co-ordinate with work.

B3 Incorrect relevant formula and continues.

B4 Transposing error.

Slips (-1)

S1 Numerical slips to a maximum of 3.

Attempts (2 marks)

A1 Any correct relevant work e.g. (0, 0) mentioned or attempt at translation or central symmetry.

Part (c)**20 (10, 10) marks****Att (3, 3)**A circle has equation $x^2 + (y-7)^2 = 100$.**(i)** Write down the co-ordinates of the centre of the circle and the radius of the circle.**(ii)** The point $(6, h)$ is on the circle. Find the two possible values of h .**c (i)****10 marks****Att 3**

$$x^2 + (y-7)^2 = 100.$$

Centre $(0, 7)$ or $h = 0, k = 7$.Radius $\sqrt{100} = 10$ **or**

$$x^2 + (y-7)^2 = 100 \Rightarrow x^2 + y^2 - 14y + 49 = 100 \Rightarrow x^2 + y^2 - 14y - 51 = 0.$$

Centre $(-g, -f) = (0, 7)$

$$\text{Radius} = \sqrt{g^2 + f^2 - c} = \sqrt{0 + 49 + 51} = 10.$$

* Accept fully correct answer without work shown.

Award only these marks in section (c)(i)

10 marks Correct centre **and** radius.3 marks Correct centre **or** radius, otherwise 0 marks.

(c) (ii)

10 marks

Att 3

$$x^2 + (y - 7)^2 = 100 \Rightarrow 6^2 + (h - 7)^2 = 100$$

$$\Rightarrow (h - 7)^2 = 100 - 36 = 64 \Rightarrow h - 7 = \pm 8$$

$$\Rightarrow h = 7 + 8 = 15 \text{ or } h = 7 - 8 = -1.$$

or

$$x^2 + (y - 7)^2 = 100 \Rightarrow 6^2 + (h - 7)^2 = 100$$

$$\Rightarrow 36 + h^2 - 14h + 49 - 100 = 0 \Rightarrow h^2 - 14h - 15 = 0 \Rightarrow (h - 15)(h + 1) = 0$$

$$\Rightarrow h = 15 \text{ or } h = -1.$$

or

$$\text{Centre} = (0, 7); \text{radius} = 10; \text{point} = (6, h) \Rightarrow \sqrt{(6 - 0)^2 + (h - 7)^2} = 10$$

$$\Rightarrow h^2 - 14h + 85 = 100$$

$$\Rightarrow h^2 - 14h - 15 = 0 \Rightarrow (h - 15)(h + 1) = 0 \Rightarrow h = 15 \text{ or } h = -1$$

* Accept (6, -1) and (6, 15) verified correctly into the equation of the circle.

Award only these marks in section (c)(i):

10 marks Both values of h fully correct from correct work.

3 marks Any work of merit, otherwise 0 marks.

Attempts (3 marks)

A1 Any work of merit, e.g. $(6)^2 = 36$.

A2 States: centre = (0, 0) or radius = 10.

A3 One or both values of h without work or only one tested.

Worthless (0)

W1 Correct formula transcribed from tables and stops.

